Dirac's theorem and geometry in the reconfiguration space of perfect matchings

Ross J. Kang* Clément Legrand-Duchesne†

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Abstract

Let G be a graph on an even number n of vertices and let $\mathcal{M}_G$ be the collection of perfect matchings in G. Dirac's theorem says that if the minimum degree $\delta(G)$ of G is at least $n/2$, then $\mathcal{M}_G$ is guaranteed to be non-empty, while this is not necessarily the case if $\delta(G) \leq n/2 - 1$.

Given an integer $k \geq 2$, let $\mathcal{H}_k(G)$ be the reconfiguration graph formed on $\mathcal{M}_G$ by connecting two distinct $M_1, M_2 \in \mathcal{M}_G$ by an edge in $\mathcal{H}_k(G)$ if M_1 can be obtained from M_2 by changing at most k edges. Besides non-emptiness, as per Dirac's theorem, what other natural properties of $\mathcal{H}_k(G)$ are guaranteed based on the minimum degree $\delta(G)$ of G?

We show that if $\delta(G) \geq \lfloor 2n/3 \rfloor + 1$, then $\mathcal{H}_2(G)$ must be connected and an expander, while for each $\delta \leq \lfloor (2n-2)/3 \rfloor$ there are graphs G with minimum degree δ such that $\mathcal{H}_2(G)$ is disconnected. We also show that, if $\delta(G) \geq n/2 + 2$, then $\mathcal{H}_3(G)$ must be connected and an expander, while for each $\delta \leq n/2 - C_k$ there are graphs G with minimum degree δ such that $\mathcal{H}_k(G)$ is disconnected, for some C_k depending on $k \geq 3$. Furthermore, for every $\varepsilon > 0$, there exists a $c > 1$ such that for every $k \geq 2$, there are graphs G with $\delta(G) = \frac{n}{2} - C_k - \varepsilon kn$ such that $\mathcal{H}_k(G)$ has at least c^n components.

With respect to guaranteeing that $\mathcal{H}_k(G)$ has positive minimum degree (or, equivalently, no isolated vertices) we show that if $\delta(G) \geq n/2 + 1$, then $\mathcal{H}_2(G)$ must have positive minimum degree. For $k \geq 3$, we show how this threshold for $\delta(G)$ is related to the notorious Caccetta-Häggkvist conjecture.

Additionally, we show analogous results for G a balanced bipartite graph on $2n$ vertices. Moreover, most of our results extend naturally to matchings of a prescribed size (instead of perfect matchings), as well as to Ore-type conditions (instead of minimum degree conditions).

1 Introduction

Let G be a graph on an even number n of vertices. We are interested in the macroscopic changes in the space of perfect matchings of G according to a natural metric, as we vary the minimum degree $\delta(G)$ of G.

It is already a fundamental result in graph theory that the space of perfect matchings undergoes some sort of phase transition at $\delta(G)$ around $n/2$. Dirac's theorem [20] states that if $\delta(G) \geq n/2$, then G must always contain a perfect matching, while if $\delta(G) < n/2$, then there are examples of G with no perfect matching. Strikingly, something suddenly occurs at this juncture. Letting $\Phi(G)$ denote the *number* of perfect matchings of G, Cuckler and Kahn [16, 17] (see also the results of Sárközy, Selkow, and Szemerédi [38] and Brègman [9]) showed that provided $\delta(G) \geq n/2$,

$$\Phi(G) \geq \left(\frac{\delta(G)}{e + o_{n\to\infty}(1)} \right)^{n/2}. \tag{1}$$

*Korteweg–de Vries Institute for Mathematics, University of Amsterdam, Netherlands (r.kang@uva.nl).

†Theoretical Computer Science Department, Faculty of Mathematics and Computer Science, Jagiellonian University, Kraków, Poland (clement.legrand-duchesne@uj.edu.pl).

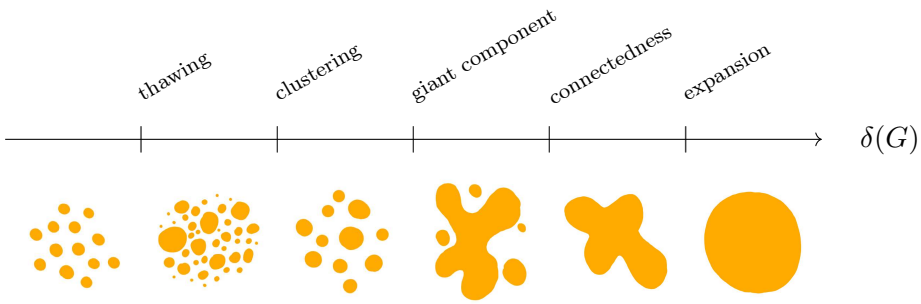


Figure 1: An artist's depiction of a possible phase diagram for the guaranteed structure of $\mathcal{H}_k(G)$ as we vary the condition on $\delta(G)$ as a function of $n = |G|$.

Given that there are so many perfect matchings, even right after the non-emptiness threshold of $\delta(G) = n/2$, a crude "volume" heuristic suggests they might "cluster".

To test this hypothesis, we consider the collection $\mathcal{M}_G$ of perfect matchings in G as a state space. We define transitions between states according to their proximity in some metric. A natural metric on $\mathcal{M}_G$ is the Hamming distance, that is, the size of the symmetric difference. Due to the definition of perfect matchings, this distance parameter can only take values in even integers at least 4. Two perfect matchings $M_1, M_2 \in \mathcal{M}_G$ have Hamming distance at most $2k$, $k \geq 2$, if and only if there is some *k-switch* between M_1 and M_2 —that is, the addition of j edges and the removal of j other edges that takes M_1 to M_2 for some $j \leq k$. Let $\mathcal{H}_k(G)$ be the state transition graph on $\mathcal{M}_G$ defined by joining two distinct $M_1, M_2 \in \mathcal{M}_G$ if M_1 and M_2 have Hamming distance at most $2k$. We call $\mathcal{H}_k(G)$ the *k-switch (reconfiguration) graph* on $\mathcal{M}_G$.

Concretely, we cast the "clustering" question above as follows. What is the smallest condition on the minimum degree $\delta(G)$ of G in terms of n that guarantees the connectedness of the k-switch graph $\mathcal{H}_k(G)$? Or a component in $\mathcal{H}_k(G)$ of size a constant proportion of $\Phi(G)$? Or positive minimum degree in $\mathcal{H}_k(G)$? Observe that if the minimum degree of G is $n-1$, then G must be the complete graph K_n and it is then easy to see that $\mathcal{H}_k(G)$ is connected. Observe also that if G is the disjoint union of $n/(2k+2)$ cycles of length $2k+2$ (for n a multiple of $2k+2$), then $\mathcal{H}_k(G)$ consists of $2^{n/(2k+2)}$ isolated vertices. Thus, by monotonicity, the above questions are all well-defined and non-trivial.

Could we obtain some "phase diagram" for $\mathcal{H}_k(G)$ (see Figure 1) similar to those observed in various random discrete systems (see for example [33, 1])? In particular, does something indeed special happen around the non-emptiness threshold at $\delta(G) = n/2$? Our main objective in this work is to sketch the first outlines for such a diagram. Our work follows in the line of other recent studies in "emergent combinatorics" [22, 7, 8, 12, 11, 23, 14] where analogous phenomena are explored in other combinatorial contexts.

Note that $\mathcal{H}_j(G) \subseteq \mathcal{H}_k(G)$ if $j \leq k$. Keeping this monotonicity in mind, we will soon see how $k = 3$ is mostly representative for any larger k. We discuss the finer subtleties later in the article, in addition to the results we have obtained for mixing time, and for matchings of some prescribed size $\gamma n/2$, where $0 \leq \gamma \leq 1$. Here are some of our main results.

Theorem 1. *Let G be a graph on an even number n of vertices. If G has minimum degree at most $n/2-2$, then the 2-switch graph $\mathcal{H}_2(G)$ may have isolated vertices. If G has minimum degree at least $n/2+1$, then $\mathcal{H}_2(G)$ has positive minimum degree.*
If G has minimum degree at most $\lfloor (2n-2)/3 \rfloor$, then $\mathcal{H}_2(G)$ may be disconnected. If G has minimum degree at least $\lfloor 2n/3 \rfloor + 1$, then $\mathcal{H}_2(G)$ is connected.

Theorem 2. *Let G be a graph on an even number n of vertices. If G has minimum degree at most $n/2-1$, then the 3-switch graph $\mathcal{H}_3(G)$ may be disconnected. If G has minimum degree at least $n/2+2$, then $\mathcal{H}_3(G)$ is connected.*

Threshold	$k=2$	$k=3$	$k \geq 4$
Thawing	$\left[\frac{n}{2}-2-o(n), \frac{n}{2}+1\right]$	$\left[\frac{n}{2}-3-o(n), \frac{n}{2}+2\right]$	$\left[\frac{n}{2}-o(n), \frac{n}{2}-k+1\right]$
Clustering	$\left[\frac{n}{2}-2-o(n), \left\lfloor\frac{2n}{3}\right\rfloor+1\right]$	$\left[\frac{n}{2}-3-o(n), \frac{n}{2}+2\right]$	$\left[\frac{n}{2}-o(kn), \frac{n}{2}-k+1\right]$
Giant component	$\left[\frac{n}{2}-2-o(1), \left\lfloor\frac{2n}{3}\right\rfloor+1\right]$	$\left[\frac{n}{2}-3-o(1), \frac{n}{2}+2\right]$	$\left[\frac{n}{2}-o(k), \frac{n}{2}-k+1\right]$
No isolated vertices	$\left[\frac{n}{2}-1, \frac{n}{2}+1\right]$	$\leq \lceil 0.3465n \rceil + 1$	Conjecture: $\frac{n}{k}+O(1)$
Connectedness	$\left[\left\lfloor\frac{2n+1}{3}\right\rfloor, \left\lfloor\frac{2n}{3}\right\rfloor+1\right]$	$\left[\frac{n}{2}, \frac{n}{2}+2\right]$	$\left[\frac{n}{2}-k, \frac{n}{2}+1\right]$
Expansion	$\left[\left\lfloor\frac{2n+1}{3}\right\rfloor, \left\lfloor\frac{2n}{3}\right\rfloor+1\right]$	$\left[\frac{n}{2}, \frac{n}{2}+2\right]$	$\left[\frac{n}{2}-k, \frac{n}{2}+1\right]$

Table 1: (general case) Bounds on thresholds for the reconfiguration geometry of perfect matchings in terms of minimum degree (see Subsection 2.3 for formal definitions of these thresholds).

Theorem 3. *Let G be a graph on an even number n of vertices. For every $\varepsilon > 1$, there exists a $c > 1$, such that the following holds for any $k \geq 2$ and n. If G has minimum degree at most $n/2 - k - \varepsilon kn$, then the k-switch graph $\mathcal{H}_k(G)$ may have c^n components and within each of these components, all matchings share a linear number of* frozen *edges.*

Informally, we can recast these results as follows (see also Table 1). If we only allow 2-switches, then the reconfiguration graph $\mathcal{H}_2(G)$ on the space $\mathcal{M}_G$ of perfect matchings in G exhibits the following phase transitions in its component structure. Until a little before the non-emptiness threshold $\delta(G) \geq n/2$, the graph $\mathcal{H}_2(G)$ can have isolated vertices and can be "shattered" into exponentially many components, each having a linear number of frozen edges in common, that is, edges belonging to all matchings of the component. Almost right after the non-emptiness threshold, $\mathcal{H}_2(G)$ must have positive minimum degree (so no isolated vertices). Markedly past the non-emptiness threshold, at around $\delta(G) = 2n/3$, there is a threshold in the connectivity of $\mathcal{H}_2(G)$: just below $2n/3$ it can be disconnected, while just above $2n/3$ the space must be connected. Note that between $n/2$ and $2n/3$ it remains open as to whether some "large" component in $\mathcal{H}_2(G)$ is guaranteed. We posit that this occurs closer to $n/2$ and pose this explicitly as an open problem (see Conjecture 3 in Section 7).

If rather than only 2-switches we allow k-switches for some fixed $k \geq 3$, then the picture narrows somewhat. Until a little before the non-emptiness threshold, the switch graph $\mathcal{H}_k(G)$ can be shattered into exponentially many components with a linear number of frozen edges. Almost right at the non-emptiness threshold, $\mathcal{H}_k(G)$ goes from potentially having no linear size component to necessarily being connected.

One of the motivations for proving that some reconfiguration graph is connected is to ensure that Markov chain Monte Carlo algorithms that sample from the corresponding space converge toward the desired probability distributions. However, connectivity alone is not enough to ensure that these algorithms are efficient: the rate of convergence of these Markov chains relates to the structural expansion properties of the reconfiguration graph. We additionally prove that $\mathcal{H}_k(G)$ becomes a weak expander right after the connectivity threshold and that one can sample in polynomial time approximately uniform random perfect matchings by iteratively applying random switches of bounded size (see Subsection 1.1 for more precise details).

In the discussion above, we had left an incomplete story with respect to positive minimum degree (or equivalently, the possible presence of isolated vertices) in $\mathcal{H}_k(G)$. Perhaps contrary to intuition, for $k \geq 3$ this threshold is more subtle and is related to a longstanding open problem.

Theorem 4. *For all $k \geq 2$, d and even n, if there is a graph G on n vertices with $\delta(G) = d$ such that $\mathcal{H}_k(G)$ contains an isolated vertex, then there exists an oriented n-vertex graph with minimum outdegree $d-1$ and no directed cycle of length at most k.*

Threshold	$k=2$	$k=3$	$k \geq 4$
Thawing	$\left[\frac{n-1}{2} - o(n), \left\lfloor \frac{n+3}{2} \right\rfloor\right]$	$\left[\frac{n-2}{2} - o(n), \left\lfloor \frac{n}{2} \right\rfloor + 1\right]$	$\left[\frac{n-k+1}{2} - o(kn), \left\lfloor \frac{n+1}{2} \right\rfloor\right]$
Clustering	$\left[\frac{n-1}{2} - o(n), \left\lfloor \frac{2n}{3} \right\rfloor + 1\right]$	$\left[\frac{n-2}{2} - o(n), \left\lfloor \frac{n}{2} \right\rfloor + 1\right]$	$\left[\frac{n-k+1}{2} - o(n), \left\lfloor \frac{n+1}{2} \right\rfloor\right]$
Giant component	$\left[\frac{n-1}{2} - o(1), \left\lfloor \frac{2n}{3} \right\rfloor + 1\right]$	$\left[\frac{n-2}{2} - o(1), \left\lfloor \frac{n}{2} \right\rfloor + 1\right]$	$\left[\frac{n-k+1}{2} - o(k), \left\lfloor \frac{n+1}{2} \right\rfloor\right]$
No isolated vertices	$\left\lfloor \frac{n+3}{2} \right\rfloor$	$\left[\left\lfloor \frac{n+5}{3} \right\rfloor, \lceil 0.3465n \rceil + 1\right]$	Conjecture: $\lceil \frac{n}{k} \rceil + 1$
Connectedness	$\left\lfloor \frac{2n}{3} \right\rfloor + 1$	$\left\lfloor \frac{n}{2} \right\rfloor + 1$	$\left[\left\lfloor \frac{n-k+1}{2} \right\rfloor, \left\lfloor \frac{n+1}{2} \right\rfloor\right]$
Expansion	$\left\lfloor \frac{2n}{3} \right\rfloor + 1$	$\left\lfloor \frac{n}{2} \right\rfloor + 1$	$\left[\left\lfloor \frac{n-k+1}{2} \right\rfloor, \left\lfloor \frac{n}{2} \right\rfloor + 1\right]$

Table 2: (bipartite case) Bounds on thresholds for the reconfiguration geometry of perfect matchings in terms of minimum degree, when G is a balanced bipartite graph on $2n$ vertices (see Subsection 2.3 for formal definitions of these thresholds).

The optimal choice of d as a function of n in the conclusion of Theorem 4 is the object of the notorious Caccetta-Häggkvist conjecture [13]. The special case $k = 3$ has drawn special attention, and the state of the art there, due to Hladký, Král', and Norin [25] using flag algebra, along with Theorem 4, implies the following.

Corollary 5. *Let G be a graph on an even number n of vertices. If G has minimum degree strictly larger than $\lceil 0.3465n \rceil$, then $\mathcal{H}_3(G)$ must have positive minimum degree.*

By Theorem 4, the hypothetical truth of the Caccetta-Häggkvist conjecture would then suggest $\lceil n/k \rceil + 1$ as an upper bound on the threshold minimum degree of G for positive minimum degree in the k-switch graph $\mathcal{H}_k(G)$. We indeed suspect that $\lceil n/k \rceil + 1$ is the correct threshold in this case and highlight this explicitly as an open problem (see Conjecture 1 in Section 7).

The special case of G being a balanced bipartite graph on $2n$ vertices, since the number of perfect matchings in this case encodes the permanent, is of particular importance. (We discuss further context for this below.) Similarly to (1), Cuckler and Kahn [16, 17] also showed in this case that provided $\delta(G) \geq n/2$,

$$\Phi(G) \geq \left(\frac{\delta(G)}{e + o_{n\to\infty}(1)} \right)^n . \tag{2}$$

We have established an almost completely analogous set of results for $\mathcal{H}_k(G)$ in this special case for G, often with slightly sharper bounds (see Table 2). An interesting point here is that we can actually establish *equivalence* between a variant of the Caccetta-Häggkvist conjecture and the minimum degree threshold that guarantees no isolated vertices in the k-switch graph of balanced bipartite graphs. Specifically, we relate this threshold to the weakening of the Caccetta-Häggkvist conjecture that constrains semidegree rather than outdegree; this version of the conjecture has also been well-studied, it being closely related to a yet older conjecture of Behzad, Chartrand, and Wall [6].

We remark here also that, both for balanced bipartite graphs and general graphs, most of our results extend to matchings of a fixed size $\gamma n/2$ for any fixed $\gamma \in (0, 1]$ by replacing $n/2$ by $\gamma n/2$ in the different thresholds. Moreover, most of our results also generalise to Ore-type conditions, where rather than mandating minimum degree, we mandate a minimum sum of the degrees of two distinct non-adjacent vertices.

As is probably already evident by now, the "volume" heuristic we offered earlier is indeed crude. If G is a balanced bipartite graph on $2n$ vertices that is moreover *d-regular*, then an important result of Schrijver [39] (see also [42] for the case $d = 3$ and [24, 15] for alternative proofs for general $d \geq 3$) gives the following lower bound on the number of perfect matchings in

G:

$$\Phi(G) \geq \left(\frac{(d-1)^{d-1}}{d^{d-2}} \right)^n .$$

This asymptotically matches the expression in (2), except that it needs no condition like $d \geq n/2$, and so *there are plenty of perfect matchings in G regardless of the value of $d \geq 3$*. On the other hand, if $d = d(n) \leq n^\theta$ for fixed $\theta < 1/(4k-1)$, it is known [35] for all sufficiently large n that there is a bipartite d-regular graph G of (finite) girth greater than $2k$ on $2n$ vertices, in which case $\mathcal{H}_k(G)$ consists of $\Phi(G)$ isolated vertices. Moreover, several of the disconnectedness constructions in Theorems 1 to 3 can be adapted so that G is regular and bipartite.

1.1 Random sampling and Markov chains on matchings

The problem of counting the number of perfect matchings of a graph attracted considerable attention over the years. When G is a balanced bipartite graph, this number is counted by the permanent of the adjacency matrix A of G:

$$\mathrm{Perm}(A) = \sum_{\sigma \in \mathfrak{S}_n} \prod_{i=1}^{n} A_{i\sigma(i)}.$$

Computing the permanent is a $\#P$-hard problem [41], which motivated the study of approximation methods. To that end, Broder [10] showed that the problem of approximately counting the number of perfect matchings can be reduced to the generation of a random perfect matching from an approximately uniform distribution. To that end, he defined a Markov chain operating on perfect and near-perfect matchings (that is matchings with at most two unmatched vertices), and showed that it mixes rapidly on balanced bipartite graphs with $2n$ vertices and minimum degree at least n. Since the ratio of near-perfect matchings to perfect matchings is bounded in these dense graphs, this also gives an efficient random approximate sampler of perfect matchings. Broder's rapid mixing proof relied on a coupling argument but contained some error and was therefore withdrawn. Fortunately, Jerrum and Sinclair [27] offered a correct proof of rapid mixing of this Markov chain (namely in time $O(n^5 \log n)$), this time relying on the canonical path method. As a result, one can sample approximately uniform perfect matchings of dense balanced bipartite graphs in time $O(n^7 \log n)$. Their proof generalises to general graphs whose ratio of near-perfect to perfect matchings is polynomial and to arbitrary balanced bipartite graphs. Under the additional assumption that the bipartite graph is γn-regular, perfect matchings can be sampled exactly uniformly, in time $O(n^{1.5+.5/\gamma} \log n)$ (see [26]).

The 2-switch Markov chain was introduced by Diaconis, Graham and Holmes [18], who proves that this Markov chain is ergodic for convex balanced bipartite graphs, that is such that all ones are consecutive in each row of the adjacency matrix. In [21], Dyer, Jerrum and Müller observe that convex balanced bipartite graphs are a subclass of chordal bipartite graphs, and prove that the 2-switch Markov chain is also ergodic in this wider class. They study a hierarchy of hereditary graphs classes: chain, monotone, biconvex, convex, chordal bipartite graphs and show that the 2-switch Markov chain mixes rapidly on monotone graphs, but exponentially on biconvex graphs.

A direct consequence of Theorem 1 (respectively Theorem 2) is that the 2-switch Markov chain (respectively the 3-switch Markov chain) could fail to be ergodic on an n-vertex graphs of minimum degree at most $\lfloor (2n-2)/3 \rfloor$ but must be ergodic if $\delta(G) \geq \lfloor 2n/3 \rfloor + 1$ (respectively at most $n/2 - 1$ and $n/2 + 2$). Furthermore, we prove that above these connectedness thresholds, the 2-switch and 3-switch Markov chains actually have polynomial mixing time.

Theorem 6. *Let G be an n-vertex graph with $\delta(G) \geq n/2+2$ (or respectively $\delta(G) \geq \lfloor 2n/3 \rfloor + 1$). Alternatively, let G be a balanced bipartite graph on $2n$ vertices with $\delta(G) \geq \lfloor n/2 \rfloor + 1$ (or*

respectively $\delta(G) \geq \lfloor 2n/3 \rfloor + 1$).
The random walk on $\mathcal{H}_3(G)$ (respectively $\mathcal{H}_2(G)$) has mixing time polynomial in n.

Although the bounds we obtain for the mixing times of the switch Markov chains are greater than the bounds of Jerrum and Sinclair on the mixing time of Broder's Markov chain, making it suboptimal for sampling purposes, the switch Markov chains have the merit of operating solely on perfect matchings.

On top of the 2-switch Markov chain and Broder's Markov chain, other Markov chains have been studied on matchings. The most famous one is arguably the monomer-dimer model that operates on all matchings and draws connections with some statistical physics model.

1.2 Organisation

We prove in Section 3 all our upper bounds on the connectedness thresholds. In Section 4, we prove the lower bounds on the connectedness, giant component, clustering and thawing thresholds for general graphs and balanced bipartite graphs. In Section 5 we prove the polynomial mixing of the switch Markov chain. Section 6 focuses on the threshold for isolated matchings. Finally, we discuss open questions and perspectives in Section 7.

2 Preliminaries

We denote $N(u)$ the neighbourhood a vertex u. Given an even cycle C, an *even chord* of C is a chord splitting the cycle in two even cycles.

2.1 Markov chains and mixing times

A Markov chain is a memoryless stochastic process over a discrete state space Ω. More precisely, a sequence of random variable $(X_0, X_1, \dots)$ is a Markov chain with transition matrix P if for all $t \in \mathbb{N}$ and $x_1, \dots x_{t+1} \in \Omega$,

$$\text{Prob}(X_{t+1} = x_{t+1} | \forall i \in [t], X_i = x_i) = \text{Prob}(X_{n+1} = x_{n+1} | X_n = x_n) = P(x_n, x_{n+1}).$$

All Markov chains considered in this article are over a finite state space Ω. By representing the distribution of X_t over Ω as a row vector μ_t with $\|\mu_t\|_1 = 1$ and i-th coordinate equal to $\text{Prob}(X_t = x_i)$, we have $\mu_t = \mu_{t-1}P$ and more generally $\mu_t = \mu_0 P^t$. With a slight abuse of notation, we will write $\mu_t(x) = \text{Prob}(X_t = x)$ and $\mu_t(A) = \text{Prob}(X_t \in A)$.

A Markov chain is *ergodic* if $\mu_t(x) > 0$ for all $x \in \Omega$, all initial distribution μ_0 and all large enough t. In particular, every ergodic Markov chain admits a unique stationary distribution π, that is such that $\pi = \pi P$. Note that π is uniform if P is symmetric. Moreover, ergodic Markov chains converge towards their stationary distribution: $\sup_{\mu_0} \left\|\mu_0 P^t - \pi\right\|_{TV}$ tends to zero as t tends to infinity, where $\|\mu - \nu\|_{TV} = \max_{A \subseteq \Omega} |\mu(A) - \nu(A)|$ is the total variation distance. The rate of this convergence is measured by the *mixing time*: $\tau_{mix} = \min\{t : d(t) < 1/4\}$, where $d(t) = \sup_{\mu_0} \left\|\mu_0 P^t - \pi\right\|_{TV}$. The constant $1/4$ might look arbitrary, but using any other constant ε instead affects the mixing time only by a factor of $\log_2(1/\varepsilon)$.

The canonical paths method

A Markov chain is said to be polynomially mixing if its mixing time is bounded by a polynomial in $\log(|\Omega|)$. The mixing time is tied to the expansion of the state space graph $\mathcal{H}$ of the Markov chain, that is the graph whose vertex set is Ω and edges corresponding to non-zero probability transitions. Several methods exist to bound the mixing time: couplings, spectral methods and the canonical paths method. We will use the last of these in this article. Let $\Gamma = (\gamma_{xy})_{x,y \in \Omega}$,

be a set of paths called *canonical paths*, joining any two states x and y of Ω in the state space graph $\mathcal{H}$. The congestion of Γ is defined as

$$\varrho(\Gamma) = \max_{(a,b)\in E(\mathcal{H})} \left\{ \frac{\sum_{(x,y)\in \mathrm{cp}(a,b)} \pi(x)\pi(y)|\gamma_{x,y}|}{\pi(a)P(a,b)} \right\}.$$

The congestion measures how evenly the set of canonical paths are distributed, and whether the resulting trace on the state space graph has good expansion, and thus bounds the mixing time:

$$\tau_{mix} \leq 2\varrho \log(|\Omega|).$$

Therefore, the core of the canonical paths methods consists in defining an appropriate set of canonical paths, and bounding its congestion.

Expansion of $\mathcal{H}_k(G)$

It is well known that rapid mixing of a random walk on a graph is tied to its expansion properties, as the main obstacle to rapid mixing are bottlenecks, and expanders graphs have bounded positive bottleneck ratio. There are many different definitions of expander graphs; we consider here the spectral one, and more specifically by considering the eigenvalues of its random-walk normalised Laplacian. Given a transition matrix P, let

$$\lambda_\star = \max\{|\lambda| \colon \lambda \text{ is an eigenvalue of } P, \lambda \neq 1\}.$$

The *spectral gap* of P is defined as $\gamma_\star = 1 - \lambda_\star$. Given some constant $c > 0$, we say that a graph is a *c-expander* if the lazy uniform random walk on $\mathcal{H}$ has spectral gap at least c. The spectral gap and the mixing time are tied as follows (see [34, Theorems 12.5 and 12.4]):

$$\Omega\left(\frac{1}{\gamma_\star}\right) \leq \tau_{\mathrm{mix}}; \tag{3}$$

$$\tau_{\mathrm{mix}} \leq O\left(\frac{\log|\mathcal{H}|}{\gamma_\star}\right). \tag{4}$$

In particular, (3) shows that a lazy random walk on a c-expander $\mathcal{H}$ mixes in time $O(\log|\mathcal{H}|/c)$. Conversely, (4) shows that in the regime of Theorem 6, $\mathcal{H}_k(G)$ is a $\Omega(n^{-d})$-expander for some fixed d. As n is the number of vertices of G, by (1), $\log|\mathcal{H}_k(G)| = \Theta(n\log n)$, hence $\mathcal{H}_k(G)$ is a $\tilde{\Omega}(1)$-expander when G satisfies the conditions of Theorem 6.

2.2 Matchings

A matching is *near-perfect* if all vertices but two are matched. A matching is called a γn-matching if exactly γn vertices are matched. Given two matchings M and M', their symmetric difference has maximum degree two. We call any cycle of $M\Delta M'$ an *alternating cycle*, any maximal path of $M\Delta M'$ an *alternating path*, and any path in $M\Delta M'$ an *alternating subpath*.

A key argument in the analysis of our Markov chain (and also in the analysis of Broder's Markov chain) is that the ratio of perfect matchings to near-perfect matchings is polynomially bounded when the minimum degree of G is sufficiently high.

Lemma 7. *Let G be an n-vertex graph in which any two non-adjacent vertices u and v have degrees that sum to at least $n-1$. Alternatively, let G be a balanced bipartite graph on $2n$ vertices such that the degrees of each pair of non-adjacent vertices u and v in different halves of the bipartition sum up to at least n. The number of near-perfect matchings of G is at most n^2 times the number of perfect matchings of G.*

Proof. Let γ be the application that maps near-perfect matchings to a couple composed of a set of two vertices and a perfect matching, defined as follows. Let N be a near-perfect matching with non-matched vertices u and v. If u and v are adjacent in G, let $\gamma(N) = (\{u, v\}, N \cup uv)$. If u and v are non-adjacent, let $V' = V(G) \setminus \{u, v\}$, let A be the subset of vertices of V' matched by N to a neighbour of v and $B = N(u) \cap V'$ [1]. We have $|A| + |B| = \deg(u) + \deg(v) > n - 2 = |V'|$ so by pigeon hole principle, there exists $x \in A \cap B$. In other words, there exists an edge xy of N such that x is adjacent to u and y to v. Let $\gamma(N) = (\{u, v\}, N \cup \{xu, yv\} \setminus xy)$.

It is straightforward to check that the application γ is injective, from which the result immediately follows. □

2.3 Thresholds in the switch graph on matchings

Here, we would like to set on more solid ground the thresholds of interest to us that we have informally referred to in Tables 1 and 2. Let $\mathcal{H}$ be a graph. (Think of $\mathcal{H}$ as some instantiation of a switch graph for matchings in some graph G.) Let $\mathcal{P}$ be a graph property. For technical reasons, we will require that $\mathcal{P}(\mathcal{H})$ always holds if $\mathcal{H}$ is the empty graph. Although there are many other $\mathcal{P}$ that we might also naturally consider, let us specify (most of) the ones we focus on in this paper. We consider $c \geq 1$ as some fixed real.

connect $\mathcal{H}$ is connected.

giant $\mathcal{H}$ has a component of size at least $|\mathcal{H}|/c$.

noiso $\mathcal{H}$ has no isolated vertices.

expand For some absolute polynomial P, the simple random walk on $\mathcal{H}$ mixes in time $P(\log |\mathcal{H}|)$ (see Subsection 2.1 for a discussion on the connections between polynomial mixing time and the expansion properties of $\mathcal{H}$).

As alluded to in the introduction, we tune the $\mathcal{H}$ under consideration by varying the minimum degree condition on the underlying graph G. In particular, we are interested in the least minimum degree for G that guarantees the corresponding $\mathcal{H}$ to have some given property $\mathcal{P}$. More precisely, given an integer $k \geq 2$ and a rational $0 \leq \gamma \leq 1$, for all n such that $\gamma n/2$ is integral and for all $\delta \in (0, n)$, let $\mathcal{H}_k(n, \gamma, \delta)$ be the following collection of graphs:

$$\{\mathcal{H}_k(G, \gamma) \colon |G| = n \text{ and } \delta(G) \geq \delta\},$$

where $\mathcal{H}_k(G, \gamma)$ denotes the k-switch reconfiguration graph on the space $\mathcal{M}_{G,\gamma}$ of $\gamma|G|$-matchings of G. Given some graph property $\mathcal{P}$ as above, we are interested in the threshold minimum degree to guarantee $\mathcal{P}$ in $\mathcal{H}_k(G, \gamma)$, defined as

$$\delta_k^{\mathcal{P}}(n, \gamma) = \min \{\delta \colon \mathcal{P}(\mathcal{H}) \text{ holds for all } \mathcal{H} \in \mathcal{H}_k(n, \gamma, \delta)\}.$$

It is straightforward to see that the minimum is well-defined if and only if $\mathcal{P}(\mathcal{H}_k(K_n, \gamma))$ holds. Note that, for the purposes of bounding threshold functions for $\mathcal{P}$, the technical condition on $\mathcal{P}$ lets us consider only those G for which $\mathcal{H}_k(G, \gamma)$ is non-empty.

Since we only consider thresholds of the above form, it makes sense to consider graph properties $\mathcal{P}$ specific to $\mathcal{H} \in \mathcal{H}_k(n, \gamma, \delta)$, that is, based possibly on properties of the underlying graph G. The followings are two such properties we consider, where again $c \geq 1$ is some fixed real. Given some edge e of the graph G and some connected subgraph $\mathcal{C}$ of $\mathcal{H}_k(G, \gamma)$, we say that e is *frozen* if $e \in M$ for all $M \in \mathcal{C}$.

[1] If G is a balanced bipartite graph, let $V' = V_1 \setminus \{v\}$, where V_1 is the half of the bipartition containing v. We then have $|A| + |B| = \deg(u) + \deg(v) > n - 1 = |V'|$.

thaw In every component of $\mathcal{H}$ (where $\mathcal{H} = \mathcal{H}_k(G, \gamma)$ for some graph G, γ), the number of non-frozen edges is at least $\gamma n/(2c)$,

cluster $\mathcal{H}$ has fewer than c^n connected components.

Note that each property $\mathcal{P}$ defined above is monotone with respect to c: if $c' > c$, then for every $\mathcal{H}$, if $\mathcal{P}_c$ is satisfied on $\mathcal{H}$ then $\mathcal{P}_{c'}$ also holds on $\mathcal{H}$. In particular, the closer c is to 1, the harder these properties are to satisfy.

Note that for convenience, we will use the abbreviated forms $\mathcal{H}_k(n, \delta)$ and $\delta_k^{\mathcal{P}}(n)$ when $\gamma = 1$. We also analogously denote by $\delta_{k,\text{bip}}^{\mathcal{P}}(n, \gamma)$ and $\delta_{k,\text{reg}}^{\mathcal{P}}(n, \gamma)$ the threshold functions where the corresponding G must be a balanced bipartite graph on $2n$ vertices and a regular balanced bipartite graph on $2n$ vertices, respectively.

For example, the inequality $\gamma n/2 - k - o(k) \leq \delta_k^{\textbf{giant}}(n, \gamma) \leq \gamma n/2 + 1$ should be read as follows. The first inequality states that for any $\varepsilon > 0$, there exists $c > 1$ such that for every integer n such that $\gamma n/2$ is integral, there exists a graph G with minimum degree at least $\gamma n/2 - k - 1 - \varepsilon k$ such that $\mathcal{H}_{k,\gamma}(G)$ contains no component of size at least $|\mathcal{H}_{k,\gamma}(G)|/c$. On the other hand, for all $c > 1$ and every n-vertex graph G with minimum degree at least $\gamma n/2 + 1$, $\mathcal{H}_{k,\gamma}(G)$ contains a component of size at least $|\mathcal{H}_{k,\gamma}(G)|/c$.

3 Upper bounds on the connectedness thresholds

In this section, we show the upper bounds on the connectedness thresholds of Theorems 1 and 2 that we recast here:

Theorem 8. *Let G be an n-vertex graph with $\delta(G) \geq n/2 + 2$ (or a balanced bipartite graph on $2n$ vertices, with $\delta(G) \geq \lfloor n/2 \rfloor + 1$). All perfect matchings of G are equivalent via 3-switches.*

Theorem 9. *Let G be an n-vertex graph with $\delta(G) \geq \lfloor 2n/3 \rfloor + 1$ (or a balanced bipartite graph on $2n$ vertices, with $\delta(G) \geq \lfloor 2n/3 \rfloor + 1$). All perfect matchings of G are equivalent via 2-switches.*

To do so, we first prove in Subsection 3.1 that perfect matchings are connected by 4-switches if $\delta \geq n/2 + 1$, before proving in Subsection 3.2 that any 4-switch can be realised by a sequence of $O(1)$ 3-switches if $\delta \geq n/2 + 2$, and in Subsection 3.3 that any 3-switch can be realised by a sequence of $O(1)$ 2-switches if $\delta \geq \lfloor 2n/3 \rfloor + 1$, which proves Theorem 8 and Theorem 9, respectively.

3.1 Equivalence via 4-switches

We first prove that perfect matchings are connected by 4-switches if $\delta \geq n/2 + 1$.

Theorem 10. *Let $\gamma \in (0, 1]$. Let G be an n-vertex graph in which each pair of non-adjacent vertices have degrees that sum to at least $\gamma n + 1$. Alternatively, let G be a balanced bipartite graph on $2n$ vertices, in which each pair of non-adjacent vertices in different halves of the bipartition have their degrees that sum up to at least γn. All γn-matchings of G are equivalent via 4-switches.*

Proof. We distinguish the case of perfect matchings from that of γn-matchings with $\gamma < 1$, the latter one being slightly more complex because of the short alternating paths.

Perfect matchings Let M and M' be two perfect matchings of G. We proceed by induction on $|M \Delta M'|$ by proving that we can always reduce the size of the symmetric difference by performing a 4-switch in M or in M'.

If there exists $C = u_1 \dots u_{2p}$ an alternating subpath of $M \Delta M'$ with $u_{2i-1}u_{2i} \in M$ and $u_{2i}u_{2i+1} \in M'$, such that $p \in \{2, 3, 4\}$ and $u_1 u_{2p} \in E(G)$, then a 4-switch in M on $u_1 \dots u_{2p}$

results in a perfect matching N such that $|N\Delta M'| \leq |M\Delta M'| - p + 1$. Thus one can assume that no such C exists.

Let $C = u_1 \dots u_6$ be an alternating subpath of $M\Delta M'$ with $u_{2i-1}u_{2i} \in M$ and $u_{2i}u_{2i+1} \in M'$. By assumption, u_1 is neither adjacent to u_4 nor to u_6, and u_6 is not adjacent to u_3. Let $V' = V(G) \setminus \{u_1, \dots u_6\}$. Let A be the subset of V' composed of vertices matched in M to a neighbour of u_1. The only vertices of $\{u_1, \dots u_6\}$ that can be matched in M to neighbours of u_1 are u_1, u_4 and u_6. Therefore, $|A| \geq \deg(u_1) - 3$. Let $B = N(u_6) \cap V'$, we also have $|B| \geq \deg(u_6) - 3$. By pigeonhole principle[2], since V' has size $n - 6$ and is a superset of A and B and $|A| + |B| \geq n - 5$, there exists a vertex x in $A \cap B$. This vertex x is by definition adjacent to u_6, and matched to a vertex y in M, itself adjacent to u_1 (see Figure 2). Performing the 4-switch $u_1 \dots u_6xy$ on M results in a perfect matching N containing u_2u_3 and u_4u_5, so $|N\Delta M'| \leq |M\Delta M'| - 1$.

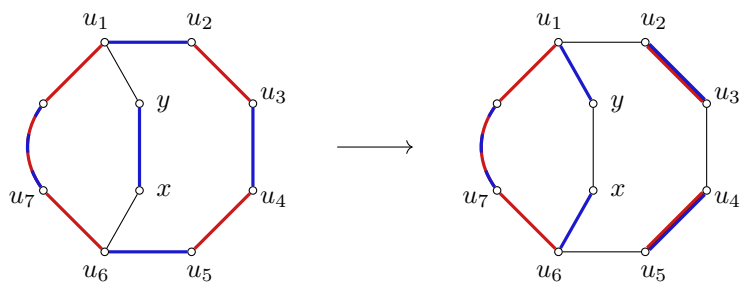


Figure 2: The reconfiguration sequence of Theorem 10. M is represented in blue and M' in red.

γn-matchings Let M and M' be two γn-matchings of G. We proceed by induction on $|M\Delta M'|$ by proving that we can always reduce the size of the symmetric difference by performing a 4-switch in M or in M'. We handle separately the following cases (in that order): isolated edges of $M\Delta M'$, even alternating paths of length at most 6, odd alternating subpaths of length 3 or 5 with an edge between their extremities, odd alternating subpaths of length 5 with no edges between their extremities and a vertex at distance 3 or 5 in the subpath, and odd alternating paths of length 3 with no edge between their extremities. It is straightforward to check that these cases are exhaustive.

If there exists an edge xy of M such that x and y are both unmatched in M', then removing any edge from $M' \setminus M$ and replacing it by xy yields a γn-matching N such that $|N\Delta M| < |M'\Delta M|$. Thus one can assume that all the edges of M are incident to edges of M' (and likewise that all edges of M' are incident to edges of M). Hence, we can assume that alternating paths have length at least two.

If there exists an even alternating path $u_1, \dots u_{2p+1}$ with $p \in [3]$, such that $u_{2i-1}u_{2i} \in M$ and $u_{2i}u_{2i+1} \in M'$, with u_1 unmatched in M' and u_{2p+1} unmatched in M, then performing the 4-switch on the path $u_1 \dots u_{2p+1}$ results in a γn-matching N with $|N\Delta M'| < |M\Delta M'|$. Thus one can assume $M\Delta M'$ contains no even alternating paths of length at most 6.

Similarly to the perfect matching case, if there exists an alternating subpath $u_1, \dots u_{2p}$ with $p \in \{2, 3, 4\}$, such that $u_{2i-1}u_{2i} \in M$ and $u_{2i}u_{2i+1} \in M'$, and $u_1u_{2p} \in E(G)$, then performing in M the 4-switch on the cycle $u_1 \dots u_{2p}$ results in a γn-matching N with $|N\Delta M'| < |M\Delta M'|$. Thus one can assume that $M\Delta M'$ contains no odd alternating subpaths of length 3 or 5 with an edge between their extremities.

Assume that there exists an odd alternating subpath $P = u_1 \dots u_6$ of length 5 in $M\Delta M'$ with $u_{2i-1}u_{2i} \in M$ and $u_{2i}u_{2i+1} \in M'$. By assumption, u_1 is neither adjacent to u_4, nor to u_6, and u_6 is not adjacent to u_3. If u_1 has a neighbour y that is unmatched in M, then

[2]If G is a balanced bipartite graph on $2n$ vertices, let $V' = V_1 \setminus \{u_1, u_3, u_5\}$, where V_1 is the half of the bipartition that contains u_1. We now have $|A| = \deg(u_1) - 1$ and $|B| = \deg(u_6) - 1$, so $|A| + |B| \geq n - 2 > |V'|$.

$N = M \cup \{yu_1, u_2u_3, u_4u_5\} \setminus \{u_1u_2, u_3u_4, u_5u_6\}$ is γn-matching such that $|N\Delta M'| \leq |M\Delta M'| - 2$. Likewise, if u_6 has a neighbour x unmatched in M, then $N = M \cup \{u_2u_3, u_4u_5, u_6x\} \setminus \{u_1u_2, u_3u_4, u_5u_6\}$ satisfies the same properties. Thus one can assume that all neighbours of u_1 and u_6 are matched in M. Let V' be the subset of vertices of $V(G) \setminus \{u_1, \dots u_6\}$ matched in M and A be the subset of V' composed of vertices matched in M to a neighbour of u_1. The only vertices of $\{u_1, \dots u_6\}$ that can be matched in M to neighbours of u_1 are u_1, u_4 and u_6. Therefore, $|A| \geq \deg(u_1) - 3$. Let $B = N(u_6) \cap V'$, we also have $|B| \geq \deg(u_6) - 3$. By pigeonhole principle[3], since V' has size $\gamma n - 6$ and is a superset of A and B and $|A| + |B| \geq \gamma n - 4$, there exists a vertex x in $A \cup B$. This vertex x is by definition adjacent to u_6, and matched to a vertex y in M, itself adjacent to u_1 (see Figure 2). Performing the 4-switch $u_1 \dots u_6xy$ on M results in a γn-matching N containing u_2u_3 and u_4u_5, so $|N\Delta M'| \leq |M\Delta M'| - 1$.

Thus, one can assume that $|M\Delta M'|$ is a disjoint union of alternating paths of length 3, each without non-adjacent extremities. Note that exactly half of them contain two edges of M. Let $P = u_1u_2u_3u_4$ and $Q = v_1v_2v_3v_4$ be two alternating paths of $M\Delta M'$, with $A = \{u_1u_2, u_3u_4, v_2v_3\} \subseteq M$ and $B = \{u_2u_3, v_1u_2, v_3v_4\} \subseteq M'$. Replacing A by B in M results in a γn-matching N with $|N\Delta M'| < |N\Delta M'|$. □

3.2 Equivalence via 3-switches

We now prove Theorem 8, which immediately follows from Theorem 10 and the following lemma.

Lemma 11. *Let G be an n-vertex graph in which each pair of non-adjacent vertices have their degrees that sum to at least $n + 3$. Alternatively, let G be a balanced bipartite graph on $2n$ vertices, in which each pair of non-adjacent vertices in different halves of the bipartition have their degrees that sum up to at least $n + 1$. Every two perfect matchings of G that differ by one 4-switch are equivalent via 3-switches.*

Proof. Let M and M' be two perfect matchings of G such that $M\Delta M'$ is a 8-cycle $C = u_1 \dots u_8$, with $u_{2i-1}u_{2i} \in M$ and $u_{2i}u_{2i+1} \in M'$. If C admits an even chords, say u_iu_j with $i < j$ and i even, then M' can be obtained from M after a switch on $u_j \dots u_8u_1 \dots u_i$, followed by a switch on $u_i \dots u_j$. One of these cycles has length 4 and the other length 6, so both are 6-switches.

Hence one can assume that C has no even chord. We first prove in Claim 12 that the existence of an edge of $M \cap M'$ with certain adjacencies implies the desired reconfiguration sequence, and secondly that such an edge exists.

Claim 12. *Let xy be an edge of $M \cap M'$ such that y is a neighbour of u_1 and x of u_2 and u_6. The matchings M and M' are equivalent via a sequence of at most three 3-switches.*

Proof of Claim. Starting from M, the sequence of 3-switches u_1u_2xy, $yu_2 \dots u_6$ and $yu_6u_7u_8u_1x$ results in M' (see Figure 3). ■

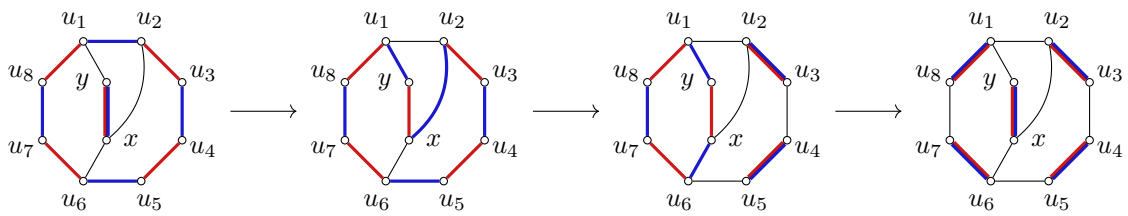


Figure 3: The reconfiguration sequence of Lemma 11. M is represented in blue and M' in red.

[3]If G is a balanced bipartite graph on $2n$ vertices, let $V' = V_1 \setminus \{u_1, u_3, u_5\}$, where V_1 is the half of the bipartition containing u_1. We now have $|A| = \deg(u_1) - 1$ and $|B| = \deg(u_6) - 1$, so $|A| + |B| \geq n - 2$.

Bipartite case For each $i \in [8]$, let E_i be the set of edges xy of $M \cap M'$ such that either x or y is adjacent to u_i. We have $|E_i| = |N(u_i) \cap (V(G) \setminus V(C))| = |N(u_i) \setminus \{u_{i-1 \pmod 8}, u_{i+1 \pmod 8}\}|$ because C contains no even chords and G is bipartite. So $|E_i| \geq \deg(u_i) - 2$. As C has no even chord, for every $i \in [8]$, u_i and $u_{i+3 \pmod 8}$ are non-adjacent and thus $\deg(u_i) + \deg(u_{i+3 \pmod 8}) \geq n + 1$. Hence,

$$\begin{aligned}\sum_{i \in [8]} |E_i| &= \sum_{i \in \{1,3,5,7\}} (\deg(u_i) + \deg(u_{i+3 \pmod 8}) - 4) \\ &\geq 4(n-3) > 4(n-4).\end{aligned}$$

Since $|M \cap M'| = n - 4$, a counting argument shows that there exists an edge $xy \in M \cap M'$ with endpoints adjacent to at least five vertices of C. Otherwise, we have $\sum_{i \in [8]} |E_i| \leq 4|M \cap M'| = 4n - 16$, a contradiction.

By the pigeonhole principle, there exists $i \in [4]$ such that u_i and u_{i+4} are adjacent to some endpoint of xy, and since G is bipartite, this endpoint is the same for u_i and u_{i+4}, say x. Without loss of generality, say that i is even. As G is bipartite and xy is adjacent to at least five vertices of C, y is adjacent to at least one vertex u_j with j odd. Up to relabelling i and j, the premise of Claim 12 is satisfied, which concludes the proof for the bipartite case.

Non-bipartite case For $i \in [8]$, let $E_i = \{(x, y) \colon xy \in M \cap M' \text{ and } x \in N(u_i)\}$. Let $V' = V(G) \setminus V(C)$. For each $i \in [8]$, we have $|E_i| = |N(u_i) \cap V'| \geq \deg(u_i) - 5$ because C has no even chords. As C has no even chord, for every $i \in [8]$, u_i and $u_{i+3 \pmod 8}$ are non-adjacent and thus $\deg(u_i) + \deg(u_{i+3 \pmod 8}) \geq n + 3$. Consider the weight function w over all pairs (x, y) such that $xy \in M \cap M'$ defined as follows: $w(x, y)$ is the number of indices $i \in [8]$ such that x is adjacent to u_i (in particular, note that $w(x, y)$ may differ from $w(x, y)$). We have by double counting:

$$\begin{aligned}\sum_{xy \in M \cap M'} w(x, y) + w(y, x) &= \sum_{(x,y) : xy \in M \cap M'} w(x, y) = \sum_{i \in [8]} |E_i| \\ &\geq \sum_{i \in \{1,3,5,7\}} (\deg(u_i) + \deg(u_{i+3 \pmod 8}) - 10) \\ &\geq 4(n-7) > 8(n/2 - 4).\end{aligned}$$

Since there are $n/2 - 4$ edges in $M \cap M'$, this implies that there exists $xy \in M \cap M'$ with $w(x, y) + w(y, x) > 8$.

For each u_i, let $w'(u_i) = |\{x, y\} \cap N(u_i)|$. We have $\sum_{i=1}^{8} w'(u_i) = w(x, y) + w(y, x) > 8$. We will shift the weights of w' towards eight targets $x_1, \ldots x_4$ and $y_1, \ldots y_4$, to prove that, up to relabelling, the adjacencies required by Claim 12 are satisfied by xy.

Consider the following discharging rules:

- For each $i \in [8]$, if u_i is adjacent x, then u_i sends weight $1/2$ to $x_{i \pmod 4}$, weight $1/4$ to $y_{i+1 \pmod 4}$ and weight $1/4$ to $y_{i+3 \pmod 4}$.
- For each $i \in [8]$, if u_i is adjacent y, then u_i sends weight $1/2$ to $y_{i \pmod 4}$, weight $1/4$ to $x_{i+1 \pmod 4}$ and weight $1/4$ to $x_{i+3 \pmod 4}$.

After applying the discharging rules, since the total weight of w' is greater than 8 and there are eight targets, one of them, say x_1 received weight greater than 1. Only four vertices (namely u_i for even i) could send weight $1/4$ to x_1, hence x_1 received a weight of $1/2$ by at least one vertex (u_1 or u_5).

If x_1 received a weight of $1/2$ by exactly one vertex, then it also received weight $1/4$ by at least three vertices, thus there exists $i \in \{2, 4\}$ such that y is adjacent to u_i and u_{i+4}. Since x is also adjacent to u_1 or u_5, the premise of Claim 12 is satisfied up to relabelling.

If x_1 received a weight of $1/2$ by two vertices, then x is adjacent to u_1 and u_5. Since x_1 also received a weight of $1/4$ by at least one vertex, y is also adjacent to a vertex u_i with i even. Up to relabelling, this is again the adjacency pattern required by Claim 12. □

3.3 Equivalence via 2-switches

Theorem 9 follows from Theorem 8 combined with the following lemma, by observing that $\frac{\lfloor 4n/3 \rfloor + 1}{2} \leq \lfloor 2n/3 \rfloor + 1$.

Lemma 13. *Let G be an n-vertex graph in which each pair of non-adjacent vertices have their degrees that sum to at least $\lfloor 4n/3 \rfloor + 1$. Alternatively, let G be a balanced bipartite graph on $2n$ vertices, in which each pair of non-adjacent vertices in different halves of the bipartition have their degrees that sum to at least $\lfloor 4n/3 \rfloor + 1$. Every two perfect matchings of G that differ by one 3-switch are equivalent via 2-switches.*

Proof. First note that for every integer d, $d \geq \lfloor (4n+3)/3 \rfloor$ if and only if $d > 4n/3$. Thus, every pair of non-adjacent vertices (in different halves of the bipartition if G is bipartite) have their degrees that sums up to more than $4n/3$. Let M and M' be two perfect matchings of G such that $M \Delta M'$ is a 6-cycle $C = u_1, \ldots u_6$ with $u_{2i-1}u_{2i} \in M$ and $u_{2i}u_{2i+1} \in M'$. Here again, if C contains an even chord $u_i u_{i+3}$ for some $i \in [3]$, then it divides C into two 4-cycles, and performing a 2-switch on each of these cycles transforms M into M'. Hence one can assume that C has no even chord. We first prove in Claim 14 that the existence of an edge of $M \cap M'$ with certain adjacencies implies the desired reconfiguration sequence, and secondly that such an edge exists.

Claim 14. *If there exists an edge $xy \in M \cap M'$ with x adjacent to u_1 and u_5, and y to u_2 and u_4, then M is equivalent to M' via a sequence of four 2-switches.*

Proof of Claim. Starting from M, the sequence of 2-switches u_1u_2yx, $yu_2u_3u_4$, $u_1xu_5u_6$ and yu_4u_5x results in M' (see Figure 4). ■

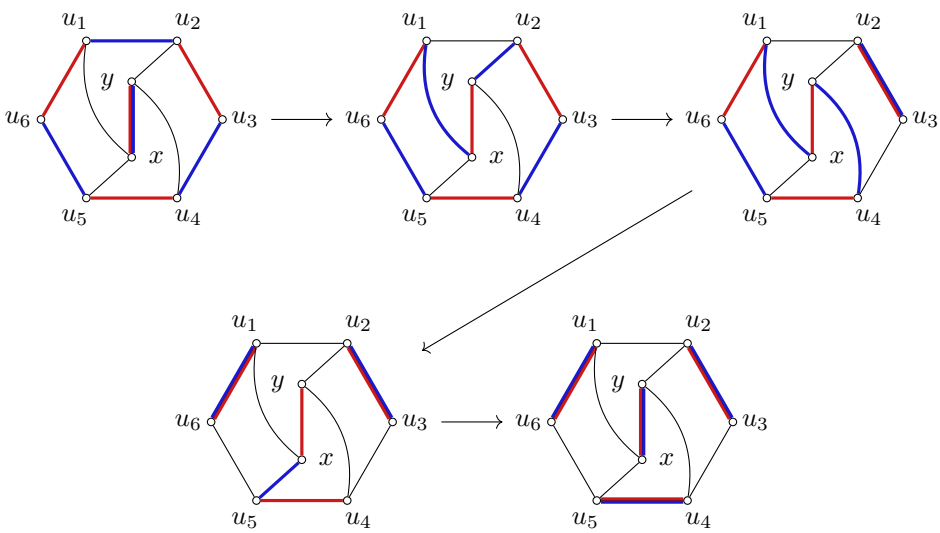


Figure 4: The reconfiguration sequence of Claim 14. M is represented in blue and M' in red.

Bipartite case For $i \in \{1, 2, 3\}$, consider E_i the set edges of $M \cap M'$ with opposite endpoints adjacent to u_i and u_{i+3} respectively. Let $V_i' = V_i \setminus V(C)$ where V_i is the half of the bipartition that contains u_i. Let A_i be the set of vertices of V_i' that are *not* matched in $M \cap M'$ to a

neighbour of u_i. We have $|A_i| = n-3-(\deg(u_i)-2) = n-1-\deg(u_i)$, because C contains no even cycle and G is bipartite so the only neighbours of u_i in $V(C)$ are u_{i+1} and $u_{i-1 \pmod 6}$. Let $B_i = V_i' \setminus N(u_{i+3})$. By the same reasoning, we also have $|B_i| = n-1-\deg(u_{i+3})$. As u_i and u_{i+3} are non-adjacent because C has no even chords, $|A_i \cup B_i| = 2n-2-\deg(u_i)-\deg(u_{i+3}) < 2n/3-2$. So for every $i \in [3]$, we have

$$\begin{aligned}|E_i| &= |V_i' \setminus (A_i \cup B_i)| \\ &> (n-3)-(2n/3-2) \\ &> n/3-1.\end{aligned}$$

As $|M \cap M'| = n-3$, by pigeon hole principle, there exists an edge xy of $M \cap M'$ that belongs to at least two of the E_i, say without loss of generality E_1 and E_2. Namely, x is adjacent to u_1 and u_5, and y to u_2 and u_4, which concludes the proof of the bipartite case by Claim 14.

Non-bipartite case For $i \in [6]$, let $E_i = \{(x,y) \colon xy \in M \cap M', x \in N(u_i), y \in N(u_{i+3 \pmod 6})\}$. Let A_i be the set of vertices of $V' = V(G) \setminus \{u_1, \dots u_6\}$ that are *not* matched to a neighbour of u_i. We have $|A_i| \leq n-6-(\deg(u_i)-4) \leq n-2-\deg(u_i)$ because C contains no even chord. Let $B_i = V' \setminus N(u_{i+3 \pmod 6})$, we also have $|B_i| \leq n-2-\deg(u_{i+3 \pmod 6})$. So for every $i \in [6]$, we have

$$\begin{aligned}|E_i| &= |V' \setminus (A_i \cup B_i)| \\ &\geq |V'| - (|A_i| + |B_i) \\ &\geq n-6-(2n-4-\deg(u_i)-\deg(u_{i+3 \pmod 6})) \\ &> n/3-2\end{aligned}$$

Consider the weight function w over all pairs (x,y) such that $xy \in M \cap M'$ defined as follows: $w(x,y)$ is the number of integers $i \in [6]$ such that x is adjacent to u_i and y to $u_{i+3 \pmod 6}$. By double counting,

$$\sum_{(x,y) \colon xy \in M \cap M'} w(x,y) = \sum_{i \in [6]} |E_i| > 6(n/3-2) = 2(n-6).$$

Since there are $n-6$ such couples (x,y), there exists (x,y) with $w(x,y) > 2$. So without loss of generality, x is adjacent to u_1 and u_5 and y to u_2 and u_4, which concludes the proof by Claim 14. □

4 Lower bounds

We first give in Subsection 4.1 general constructions yielding lower bounds on the thawing, clustering, giant-component and connectedness thresholds for any k. In Subsection 4.2 we give constructions yielding sharper bounds for the connectedness thresholds when $k \in \{2,3\}$, as well as lower bounds for the connectedness threshold of regular balanced bipartite graph.

4.1 Lower bounds for generic k

We first define two families of graphs, one composed of balanced bipartite graphs, the other of general graphs, that we will use for several of our lower bounds. For every $\gamma \in (0,1]$, for every integers $k \geq 2$, $p \geq 1$ and n such that $\gamma n/2$ is integral and at least $p(k+1)$, let $G_{k,p,\gamma,n}$ be the n-vertex graph constructed as follows. We partition the vertex set $V(G_{k,p,\gamma,n})$ into three sets $X \sqcup Y \sqcup Z$ with $|X| = 2p(k+1)$, $|Y| = \gamma n/2 - p(k+1)$ and $|Z| = (1-\gamma/2)n - p(k+1)$. Note that these quantities are integral because we assumed that $\gamma n/2$ is integral so that the graph could have a matching of size exactly $\gamma n/2$. We add edges such that the set X induces a collection of

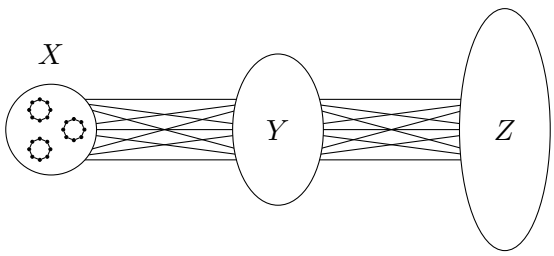


Figure 5: The graph $G_{k,p,\gamma,n}$, here with $p = 3$ and $k = 3$.

p vertex-disjoint cycles of length $2(k+1)$, Y and Z are independent sets, and all edges between Y and $X \cup Z$ are present (see Figure 5).

Lemma 15. *Let $\gamma \in (0,1]$ and k, p, n be integers with $k \geq 2$, $p \geq 1$ and $\gamma n/2 \geq p(k+1)$ an integer. The graph $G_{k,p,\gamma,n}$ has minimum degree $\gamma n/2 - p(k+1)$ and under k-switches, the space of γn-matchings of $G_{k,p,\gamma,n}$ has exactly 2^p connected components, which have equal size. Moreover, the restrictions of the γn-matchings to X are constant within each of these connected components, i.e. $p(k+1)$ frozen edges.*

Proof. The vertices in Y have degree $(1-\gamma/2)n + p(k+1) \geq n/2$, the vertices of X have degree larger than those in Z, which have degree $\gamma n/2 - p(k+1)$, so $\delta(G_{k,p,\gamma,n}) = \gamma n/2 - p(k+1)$.

Let M be a γn-matching of $G_{k,p,\gamma,n}$. The number of edges in M is $\gamma n/2 = |Y| + |X|/2$. Since Z and Y are independent sets with a complete bipartite graph between them, M has $|Y|$ edges connecting Y to Z and $|X|/2$ edges within X. In other words, M induces a perfect matching on X, and all vertices of Y are matched to a vertex of Z. As $G_{k,p,\gamma,n}[X]$ is a collection of p cycles of length $2(k+1)$, it has exactly 2^p perfect matchings. Any two of these matchings differ on at least one cycle, that is on $2(k+1)$ edges, which proves that $G_{k,p,\gamma,n}[X]$ (and respectively $G_{k,p,\gamma,n}$) have at least 2^p equivalence classes of γn-matchings under k-switches. As $G_{k,p,\gamma,n}[Y \cup Z]$ is a bipartite complete graph, all matchings of $G_{k,p,\gamma,n}[Y \cup Z]$ saturating Y are equivalent under 2-switches. Therefore, the equivalence classes of the γn-matchings of $G_{k,p,\gamma,n}$ under k-switches are in one to one correspondence with the restriction of the matchings to X, which proves that the number of equivalence classes is exactly 2^p and that the restrictions of the γn-matchings to X are constant within each of the equivalence classes. □

We now construct a balanced bipartite graph similar to $G_{k,p,\gamma,n}$. For all $\gamma \in (0,1]$, for all integer $k \geq 2$, $p \geq 1$ and n such that γn is integral and at least $p(k+1)$, let $G^{(\mathcal{B})}_{k,p,\gamma,n}$ be the balanced bipartite graph on $2n$ vertices constructed as follows. Denote $V_1 \sqcup V_2$ the bipartition of $V(G^{(\mathcal{B})}_{k,p,\gamma,n})$. For each i, we partition V_i in three sets $X_i \sqcup Y_i \sqcup Z_i$ with $|X_i| = p(k+1)$, $|Y_1| = \lceil \gamma n/2 - p(k+1)/2 \rceil$ and $|Z_1| = \lfloor (1-\gamma/2)n - p(k+1)/2 \rfloor$, and Y_2 and Z_2 of size like Y_1 and Z_1 with the floors and ceilings permuted. For each $i \in [2]$ we add all the edges between Y_i and $X_{3-i} \cup Z_{3-i}$. Finally, $G^{(\mathcal{B})}_{k,p,\gamma,n}[X_i \cup X_2]$ is a disjoint collection p cycles of length $2(k+1)$ alternating between X_1 and X_2 (see Figure 6).

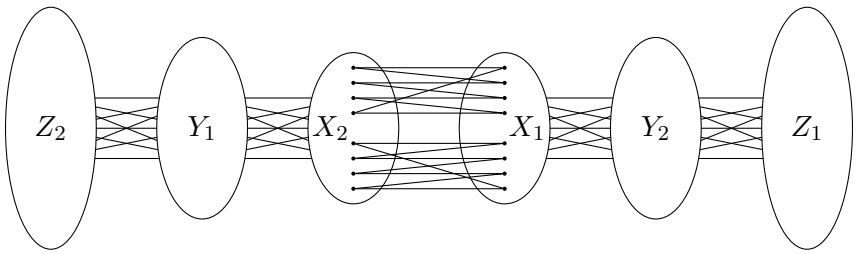


Figure 6: The graph $G^{(\mathcal{B})}_{k,p,\gamma,n}$, here with $p = 2$ and $k = 3$.

Lemma 16. *Let $\gamma \in (0,1]$ and k,p,n be integers with $k \geq 2$, $p \geq 1$ and $\gamma n \geq p(k+1)$ an integer. The graph $G^{(\mathcal{B})}_{k,p,\gamma,n}$ has minimum degree $\lfloor(\gamma n - p(k+1))/2\rfloor$ and under k-switches, the space of γn-matchings of $G^{(\mathcal{B})}_{k,p,\gamma,n}$ has exactly 2^p connected components, which have equal size. Moreover, the restrictions of the γn-matchings to $X_1 \cup X_2$ are constant within each of these connected components, i.e. $p(k+1)$ frozen edges.*

Proof. This proof is highly similar to that of Lemma 15. The vertices of Y_i have degree at least $p(k+1) + \lfloor(1-\gamma/2)n - p(k+1)/2\rfloor \geq n/2$, the vertices of X_i have degree at least $\lfloor\gamma n - p(k+1)/2\rfloor + 2$ and the vertices of Z_i at least $\lfloor\gamma n - p(k+1)/2\rfloor$, so $\delta(G^{(\mathcal{B})}_{k,p,\gamma,n}) = \lfloor\gamma n - p(k+1)/2\rfloor$.

Let M be a perfect matching of $G^{(\mathcal{B})}_{k,p,\gamma,n}$. The number of edges in M is $\gamma n = |Y_1| + |Y_2| + |X_1 \cup X_2|/2$, and as the vertices in $Z_1 \cup Z_2$ can only be matched to $Y_1 \cup Y_2$, M induces a perfect matching on $X_1 \cup X_2$ and all vertices of $Y_1 \cup Y_2$ are matched to vertices of $Z_1 \cup Z_2$. As $X_1 \cup X_2$ induces a collection of p cycles of length $2(k+1)$, $G^{(\mathcal{B})}_{k,p,\gamma,n}[X_1 \cup X_2]$ has exactly 2^p perfect matchings. Any two of these matchings differ on at least one cycle, that is on $2(k+1)$ edges, which proves that $G^{(\mathcal{B})}_{k,p,\gamma,n}[X_1 \cup X_2]$ (and respectively $G^{(\mathcal{B})}_{k,p,\gamma,n}$) have at least 2^p equivalence classes of γn-matchings under k-switches. As $G^{(\mathcal{B})}_{k,p,\gamma,n}[Y_1 \cup Z_2]$ and $G^{(\mathcal{B})}_{k,p,\gamma,n}[Y_2 \cup Z_1]$ are bipartite complete graphs, all matchings of $G^{(\mathcal{B})}_{k,p,\gamma,n}[Y_1 \cup Y_2 \cup Z_1 \cup Z_2]$ saturating $Y_1 \cup Y_2$ are equivalent under 2-switches. Therefore, the equivalence classes of the γn-matchings of $G^{(\mathcal{B})}_{k,p,\gamma,n}$ under k-switches are in one to one correspondence with the restriction of the matchings to $X_1 \cup X_2$, which proves that the number of equivalence classes is exactly 2^p and that the restrictions of the γn-matchings to $X_1 \cup X_2$ are constant within each of the equivalence classes. □

As a result, we have the following three corollaries by adjusting the value of p in $G_{k,p,\gamma,n}$ and $G^{(\mathcal{B})}_{k,p,\gamma,n}$:

Corollary 17. *For all $\varepsilon > 0$, there exists $c > 1$ such that for all $k \geq 2$, for all $\gamma \in (0,1]$ and for all n such that $\gamma n/2$ is integral, there exists an n-vertex graph G with minimum degree at least $\gamma n/2 - \varepsilon kn$ such that $\mathcal{H}_{k,\gamma}(G)$ has c^n connected components (alternatively, G is a balanced bipartite graph on $2n$ vertices, with the same minimum degree and the same condition on $\mathcal{H}_{k,\gamma}(G)$). In other words, we have the following inequalities:*

$$\delta_k^{\textbf{cluster}}(n,\gamma) \geq \frac{\gamma n}{2} - k - o(kn), \quad \text{and} \quad \delta_{k,\text{bip}}^{\textbf{cluster}}(n,\gamma) \geq \left\lfloor \frac{\gamma n - (k-1)}{2} \right\rfloor - o(kn).$$

Corollary 18. *For all $\varepsilon > 0$, there exists $c > 1$ such that for all $k \geq 2$, for all $\gamma \in (0,1]$ and for all n such that $\gamma n/2$ is integral, there exists an n-vertex graph G with minimum degree at least $\gamma n/2 - \varepsilon n$ such that each connected component of $\mathcal{H}_{k,\gamma}(G)$ has at most $\gamma n/(2c)$ non-frozen edges (alternatively, G is a balanced bipartite graph on $2n$ vertices, with the same minimum degree and the same condition on $\mathcal{H}_{k,\gamma}(G)$).*
In other words, we have the following inequalities:

$$\delta_k^{\textbf{thaw}}(n,\gamma) \geq \frac{\gamma n}{2} - k - o(n), \quad \text{and} \quad \delta_{k,\text{bip}}^{\textbf{thaw}}(n,\gamma) \geq \left\lfloor \frac{\gamma n - (k-1)}{2} \right\rfloor - o(n).$$

Corollary 19. *For all $\varepsilon > 0$, there exists $c > 1$ such that for all $k \geq 2$, for all $\gamma \in (0,1]$ and for all n such that $\gamma n/2$ is integral, there exists an n-vertex graph G with minimum degree at least $\gamma n/2 - (1+\varepsilon)(k+1)$ such that all connected components of $\mathcal{H}_{k,\gamma}(G)$ have size less than $|\mathcal{H}_{k,\gamma}(G)|/c$ (alternatively, G is a balanced bipartite graph on $2n$ vertices, with minimum degree at least $\left\lfloor \frac{\gamma n - (1+\varepsilon)(k+1)}{2} \right\rfloor$ and the same condition on $\mathcal{H}_{k,\gamma}(G)$).*
In other words, we have the following inequalities:

$$\delta_k^{\textbf{giant}}(n,\gamma) \geq \frac{\gamma n}{2} - (1+o(1))k, \quad \text{and} \quad \delta_{k,\text{bip}}^{\textbf{giant}}(n,\gamma) \geq \left\lfloor \frac{\gamma n - (1+o(1))(k-1)}{2} \right\rfloor.$$

Corollary 20. *For all $k \geq 2$, for all $\gamma \in (0,1]$ and for all n such that $\gamma n/2$ is integral, there exists an n-vertex graph G with minimum degree at least $\gamma n/2 - (k+1)$ such that $\mathcal{H}_{k,\gamma}(G)$ is disconnected (alternatively, G is a balanced bipartite graph on $2n$ vertices, with minimum degree at least $\left\lfloor \frac{\gamma n-(k+1)}{2} \right\rfloor$ and the same condition on $\mathcal{H}_{k,\gamma}(G)$).*
In other words, we have the following inequalities:

$$\delta_k^{\textbf{connect}}(n,\gamma) \geq \frac{\gamma n}{2} - k \quad \text{and} \quad \delta_{k,\text{bip}}^{\textbf{connect}}(n,\gamma) \geq \left\lfloor \frac{\gamma n - (k-1)}{2} \right\rfloor.$$

Proofs of Corollaries 17 to 20. For the clustering threshold, let $0 < \varepsilon < \gamma(k+1)$ and let $p = 1 + \lfloor \varepsilon n/2 \rfloor$. We have $p(k+1) \leq \varepsilon n(k+1)/2 \leq \gamma n/2$. Thus, $G_{k,p,\gamma,n}$ has minimum degree $\gamma n/2 - p(k+1) > \gamma n/2 - (k+1) - \varepsilon kn$ and by Lemma 15, $\mathcal{H}_{k,\gamma}(G_{k,p,\gamma,n})$ has $2^p \geq c^n$ connected components, where $c = 2^\varepsilon$.

For the thawing threshold, let $0 < \varepsilon < 1$ and let $p = 1 + \lfloor \varepsilon\gamma n/2(k+1) \rfloor$. We have $p(k+1) \leq \varepsilon\gamma n/2$. Thus, $G_{k,p,\gamma,n}$ has minimum degree $\gamma n/2 - p(k+1) > \gamma n/2 - (k+1) - \varepsilon n$ and by Lemma 15, $\mathcal{H}_{k,\gamma}(G_{k,p,\gamma,n})$ has 2^p connected components, each with $p(k+1)$ frozen edges, and thus at most $\gamma n/2 - p(k+1) \leq \frac{\gamma n}{2c}$ non-frozen edges, where $c = 1/(1-\varepsilon)$.

For the giant component threshold, let $0 < \varepsilon$ and let $p = 1 + \lfloor \varepsilon \rfloor$. We have $p(k+1) \leq (1+\varepsilon)(k+1)$. Thus, $G_{k,p,\gamma,n}$ has minimum degree $\gamma n/2 - p(k+1) > \gamma n/2 - (1+\varepsilon)(k+1)$ and by Lemma 15, $\mathcal{H}_{k,\gamma}(G_{k,p,\gamma,n})$ has 2^p connected components of equal size, hence all of them have size at most $|\mathcal{H}_{k,\gamma}(G_{k,p,\gamma,n})|/c$ where $c = 2^p = 2^{1+\lfloor \varepsilon \rfloor}$.

For the connectedness threshold, let $p = 1$. Then, $G_{k,p,\gamma,n}$ has minimum degree $\gamma n/2 - (k+1)$ and by Lemma 15, $\mathcal{H}_{k,\gamma}(G_{k,p,\gamma,n})$ has 2 connected components.

We obtain the bounds for balanced bipartite graphs by the same arguments. □

4.2 Precise bounds for non-connectedness and regular bipartite graphs

We construct two families of graphs, one composed of balanced bipartite (regular) graphs, the other of general graphs. For any $k \geq 2$ and any even n, let $F_{k,n}$ be the graph constructed as follows. First assume that n is a multiple of $(k+1)$ to avoid divisibility issues. We partition the vertex set $V(F_{k,n})$ in $(k+1)$ sets $\bigsqcup_{i\in[k+1]} X_i$ such that $|X_i| = n/(k+1)$. The sets X_1 and X_{k+1} each induce a complete graph, all other X_i induce an independent set and $F_{k,n}[X_i \cup X_{i+1}]$ induces a complete bipartite graph for each $i \in [k]$ (Figure 7). Now, if n is not a multiple of $k+1$, we add the remaining $n \pmod{k+1}$ vertices in different set X_i, starting by X_2, then X_k (if $k \neq 2$), and then X_1 and X_{k+1}. In particular, for $k = 2$ and $k = 3$, $\delta(F_{k,n})$ is attained by the vertices of X_{k+1}, thus $\delta(F_{2,n}) = \lceil \frac{n}{3} \rceil + \lfloor \frac{n}{3} \rfloor - 1 = \lfloor \frac{2n-2}{3} \rfloor$ and $\delta(F_{3,n}) = \lfloor \frac{n-2}{4} \rfloor + \lfloor \frac{n}{4} \rfloor = \frac{n}{2} - 1$ because n is even.

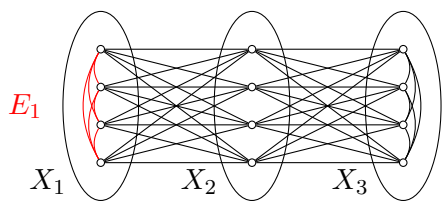


Figure 7: The graph $F_{2,n}$. The edges in the set E_1 (defined in the proof of Lemma 21) are drawn in red.

Lemma 21. *Let $k \geq 2$ and $n \geq 4(k+1)$ be an even integer. The switch graph $\mathcal{H}_k(F_{k,n})$ is disconnected.*

Proof. Let E_1 be the subset of edges with both endpoints in X_1 and E_2 the remaining edges. The number of matched edges of E_1 in a perfect matching is invariant under k-switches. Indeed,

let M be a perfect of $F_{k,n}$ and C an alternating cycle of M of length at most $2k$ containing some edge of X_1. The distance between X_1 and X_{k+1} is k, so C does not intersect X_{k+1}. Hence, C can be decomposed as the concatenation of paths $P_1, Q_1, \dots P_\ell, Q_\ell$, where the paths P_i use only edges of E_1 and the paths Q_i use only edges of E_2. In particular, each Q_i has even length, because the subgraph on the vertex set $V(F_{k,n}) \setminus X_{k+1}$ containing the edges E_2 is bipartite. So performing a switch on the alternating cycle C preserves the number of edges in E_2 and thus the number of edges in E_1.

There exists a perfect matching with exactly p edges of E_1 for each $p \in \{1, \dots \lfloor n/(2(k+1))\rfloor\}$, so $\mathcal{H}_k(F_{k,n})$ is disconnected because $n \geq 4(k+1)$. □

We now construct an analogous balanced bipartite graph. Let $k \geq 2$ and $n \geq 2(k+1)$ be two integers. Assume for now that n is a multiple of $(k+1)$. Let $F^{(\mathcal{B})}_{k,n}$ be the balanced bipartite graph on $2n$ vertices obtained by replacing each vertex of a $2(k+1)$ cycle by an independent set of size $n/(k+1)$ (see Figure 7). Note that $F^{(\mathcal{B})}_{k,n}$ is $2n/(k+1)$-regular.

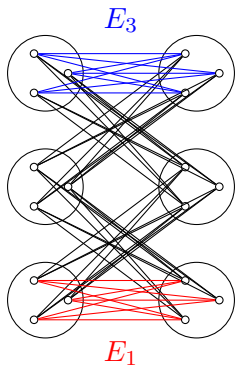


Figure 8: The graph $F^{(\mathcal{R})}_{2,n}$, i.e. a blow-up of the vertices of a 6-cycle into independent set of size $n/3$. The edges in the sets E_1 and E_3 (defined in the proof of Lemma 22) are drawn in red and blue respectively.

Now if n is not a multiple of $(k+1)$, we construct $F^{(\mathcal{B})}_{k,n}$ similarly, by distributing the additional vertices one-by-one in pairs of consecutive blown-up vertices. The graph we obtain is not regular anymore, but still balanced bipartite and we have $\delta(F_{2,n}) = \lfloor 2n/3 \rfloor$ and $\delta(F_{3,n}) = \lfloor n/2 \rfloor$.

Lemma 22. *Let $k \geq 2$ and $n \geq (k+1)$ an integer. The switch graph $\mathcal{H}_k(F^{(\mathcal{B})}_{k,n})$ is disconnected.*

Proof. This proof is highly similar to that of Lemma 21. Denote $u_1 \dots u_{2(k+1)}$ the vertices of $C_{2(k+1)}$. In $F^{(\mathcal{B})}_{k,n}$, denote E_1 the set of edges joining the blow-ups of u_1 and u_2, E_3 the set of edges joining the blow-ups of u_{k+1} and u_{k+2}, and E_2 the set of remaining edges. As in Lemma 21, the number of matched edges of E_1 in a perfect matching is invariant under k-switches. Indeed, any cycle of length at most $2k$ cannot simultaneously contain an edge of E_1 and an edge of E_3. Thus an alternating cycle C of length at most $2k$ containing an edge of $M \cap E_1$ can be decomposed as the concatenation of paths P_1, $Q_1 \dots P_\ell$, Q_ℓ where the paths P_i use only edges of E_1 and the paths Q_i only edges of E_2. In particular, as the edges of E_2 form a bipartite graph, the paths Q_i are of even length and a switch on C preserves the number of edges in E_2 and thus the number of edges in E_1. For each $p \in \{0, \dots \lfloor n/(k+1)\rfloor\}$, there exists a perfect matching with exactly p edges of E_1, so $\mathcal{H}_k(F^{(\mathcal{R})}_{k,n})$ is disconnected. □

Lemmas 21 and 22 directly give us tighter lower bounds than Lemmas 15 and 16 on the connected thresholds for $k \in \{2, 3\}$, as well as a lower bound on the connectedness threshold for balanced bipartite regular graphs.

Corollary 23. *For n large enough, we have the following inequalities.*

- $\delta_2^{\mathbf{connect}}(n) \geq \left\lfloor \frac{2n+1}{3} \right\rfloor$.
- $\delta_3^{\mathbf{connect}}(n) \geq \frac{n}{2}$.
- $\delta_{2,\text{bip}}^{\mathbf{connect}}(n) = \lfloor \frac{2n}{3} \rfloor + 1$.
- $\delta_{3,\text{bip}}^{\mathbf{connect}}(n) = \lfloor \frac{n}{2} \rfloor + 1$.
- *For all $k \geq 2$, if $(k+1)$ divides n, then $\delta_{k,\text{reg}}^{\mathbf{connect}}(n) \geq 2\frac{n}{k+1} + 1$.*

5 Random sampling

Similarly as for our connectedness results, we first prove in Subsection 5.1 that the 4-switch Markov chain mixes polynomially, before deriving from it the polynomial mixing of the 2-switch and 3-switch Markov chain in Subsection 5.2.

5.1 Polynomial mixing time of the 4-switch Markov chain

In the rest of this section, G is either an n-vertex graph or a balanced bipartite graph on $2n$ vertices. Consider the following Markov Chain Γ on the perfect matchings of G. Given a perfect matching M of G:

1. Select independently and uniformly at random four vertices u_1, u_2, u_3 and u_4, (if G is balanced bipartite, select all them from the same side of the bipartition). Let $u_i v_i$ be the edge of M for each i.
2. If $u_l = u_1$ for some $l \in \{2, 3, 4\}$, let l be the smallest such index and $C = (u_1, v_1, u_2, v_2 \ldots u_{l-1}, v_{l-1})$. Otherwise, let C be the sequence $(u_1, v_1, u_2, v_2, u_3, v_3, u_4, v_4)$.
3. If C is a simple cycle, set $M' \leftarrow M \Delta C$ with probability $1/2$ and $M' \leftarrow M$ with probability $1/2$.

Note that the probability of performing a 4-switch on a fixed cycle C of length $\ell \in \{4, 6, 8\}$ is $\Theta(1/n^{\ell/2})$. Moreover, the transition matrix of Γ is symmetric, so Γ admits the uniform distribution on the space of perfect matchings of G as a stationary distribution. The probability of not performing any switch is at least $1/2$ because of laziness of the third step, hence Γ is also aperiodic. Hence, by Theorem 10, Γ is ergodic and converges towards the uniform distribution on perfect matchings if G has sufficiently high minimum degree:

Theorem 24. *Let G be an n-vertex graph in which the degrees of each pair of non-adjacent vertices sum up to at least $n+2$. Alternatively, let G be a balanced bipartite graph on $2n$ vertices in which each pair of non-adjacent vertices in different halves of the bipartition have their degrees that sum up to at least $n+1$.*
The Markov chain Γ on the perfect matchings of G converges towards the uniform distribution on the set of perfect matchings of G. It has mixing time

$$\tau_{mix} \leq O(n^8 \log(n)).$$

The proof of the mixing time of Theorem 24 is a textbook application of the canonical paths method. The canonical paths we will use are close to the reconfiguration sequences used to prove Theorem 10, but operate on the symmetric difference between the endpoints of the path following a precise order.

Proof. To bound the congestion of our Markov chain Γ, we will define a canonical path $\gamma_{S,T}$ connecting each pair of perfect matchings S and T. Consider an arbitrary order on the vertices of G. Given S and T two perfect matchings of G, our canonical path between S and T is defined as follows. Let $C_1, \dots C_p$ be the alternating cycles in $S\Delta T$, such that for each i, the smallest vertex in C_i is smaller than the smallest vertex in C_{i+1}. Let $M_1, \dots M_{p+1}$ be the perfect matchings of G such that M_i is equal to T on all cycles C_j with $j < i$, equal to S on all cycles C_j with $j \geq i$ and equal to S (and T) around all remaining vertices. Our canonical path will go from $S = M_1$ to $T = M_{p+1}$ and will visit all M_i in increasing order. Thus we only need to explain how to go from M_i to M_{i+1}.

Connecting M_i to M_{i+1}

Denote $v_1, v_2, \dots v_{2q}$ the consecutive vertices of C_i, such that v_1 is the smallest vertex and $v_1v_2 \in M_i$.

Claim 25. *There exists a sequence $M_i = N_0, \dots N_q = M_{i+1}$ of perfect matchings of G such that for all j, N_{j-1} and N_j differ by a 4-switch and one of the following conditions is verified:*

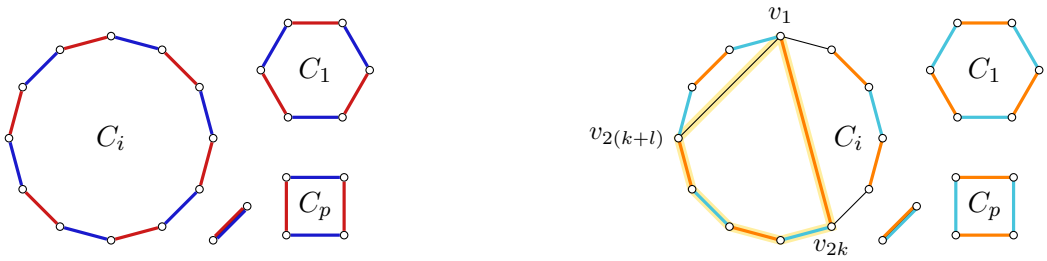
- *There exists $k \in [q]$ such that N_j is equal to M_i on $G \setminus C_i$ and contains the edges $v_1v_{2k}, v_{2k+1}v_{2k+2}, \dots v_{2q-1}v_{2q}$ and $v_2v_3, \dots v_{2k-2}v_{2k-1}$. In other words, N_j and M_i are equal on all vertices but $\{v_1, \dots v_{2k}\}$, where N_j contains v_1v_{2k} and is equal to M_{i+1} on all other vertices. We will say that N_j is representative and call k its progress index.*
- *There exists $k \in [q]$ and $xy \in N_{j-1}$ such that N_j is equal to M_i on $G \setminus (C_i \cup \{x, y\})$ and contains the edges $v_1x, yv_{2k}, v_{2k+1}v_{2k+2}, \dots v_{2q-1}v_{2q}$ and $v_2v_3, \dots v_{2k-2}v_{2k-1}\}$. In other words, N_j and M_i are equal on all vertices but $\{v_1, \dots v_{2k}, x, y\}$, where N_j contains v_1x and yv_{2k} and is equal to M_{i+1} on all other vertices. Moreover, neither v_{2k} nor v_{2k+2} are adjacent to v_1. We will call such N_j misleading, k its progress index xy the deceptive edge.*

Proof of Claim. We build the sequence $N_0, \dots N_q$ inductively. Suppose we already constructed $N_0, \dots N_{j-1}$. We distinguish several cases, depending on whether N_{j-1} and the matching N_j we construct are representative or misleading.

Case 1 First, assume that N_{j-1} is representative, let k be the progress index of N_{j-1} and assume that there exists some $l \in [3]$ such that $v_{2(k+l)}$ is a neighbour of v_1. Then let l be the largest such integer. Let N_j be the perfect matching obtained by performing the 4-switch on $C = v_1v_{2k}v_{2k+1} \dots v_{2(k+l)}$. The vertices affected by this 4-switch all belong to C_i. Since N_{j-1} was representative and N_j contains the edges $v_1v_{2(k+l)}$, $v_{2(k+l)+1}v_{2(k+l)+2}$, $\dots v_{2q-1}v_{2q}$ and v_2v_3, $\dots v_{2(k+l)-2}v_{2(k+l)-1}$, N_j is also representative with progress index $k + l$ (see Figure 9).

Case 2 Assume now no such l exists, but N_{j-1} is still representative. Let A be the subset of vertices of $V' = V(G) \setminus \{v_1, v_{2k}, v_{2k+1}, \dots v_{2k+4}\}$ matched in N_{j-1} to a neighbour of v_1. Since v_{2k+2} and v_{2k+4} are not neighbours of v_1, we have $|A| \geq \deg(v_1) - 3$[4]. Likewise, v_{2k+4} is not a neighbour of v_1 so $B = N(v_{2k+4}) \cap V'$ has size at least $\deg(v_{2k+4}) - 4$. We have $|V'| = n - 6$ and $|A| + |B| \geq n - 5$ because v_1 and v_{2k+4} are not adjacent. So by pigeonhole principle, there exists $x'y' \in N_{j-1}$, with $x', y' \notin \{v_1, v_{2k}, \dots v_{2k+4}\}$, such that x' is a neighbour of v_1 and y' of v_{2k+4}. By performing the 4-switch $C = v_1v_{2k} \dots v_{2k+4}y'x'$ in N_{j-1}, one obtains the misleading matching N_j with the deceptive edge $x'y'$ and progress index $k + 2$ (see Figure 10). Moreover, note that since l did not exist, neither v_{2k+4} nor v_{2k+6} is adjacent to v_1.

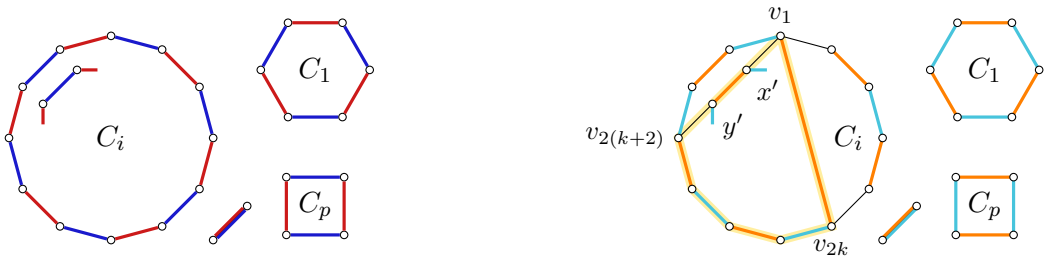
[4] If G is balanced bipartite, we set $V' = V_1 \setminus \{v_1, v_{2k+1}, v_{2k+3}\}$ where V_1 is the half of the bipartition containing v_1. We then have $|A| \geq \deg(v_1) - 1$. We also have $|B| = |N(v_{2k+4}) \cap V'| \geq \deg(v_{2k+4}) - 3$ because v_1 and v_{2k+4} are not adjacent. Since $|V'| = n - 3$ and $|A| + |B| \geq n - 2$, by pigeonhole principle, A and B intersect.



(a) The perfect matchings S, and T are represented in blue and red respectively.

(b) The matchings N_{j-1} and L (defined in Claim 26), represented in orange and light blue respectively. N_j is obtained by performing a 4-switch on C, which is here highlighted in yellow.

Figure 9: Constructing N_j when N_{j-1} is representative and v_1 is a neighbour of $v_{2(k+l)}$ for some $l \in [3]$ (here $l = 2$). The resulting N_j is also representative.



(a) The perfect matchings S, and T are represented in blue and red respectively.

(b) The matchings N_{j-1} and L (defined in Claim 26), represented in orange and light blue respectively. N_j is obtained by performing a 4-switch on C, which is here highlighted in yellow.

Figure 10: Constructing N_j when N_{j-1} is representative and v_1 has no neighbours among $v_{2(k+1)}$, $v_{2(k+2)}$ and $v_{2(k+3)}$. The resulting N_j is misleading.

Setup for the remaining cases Now, assume that N_{j-1} is misleading. Let xy be the deceptive edge of N_{j-1} and k the progress index of N_{j-1}. We distinguish cases depending on the value of (x, y): the most generic case is when $\{x, y\} \cap \{v_{2k+1}, \dots v_{2k+4}\} = \emptyset$, otherwise $\{x, y\} = \{v_{2k+1}, v_{2k+2}\}$ or $\{x, y\} = \{v_{2k+3}, v_{2k+4}\}$ because xy is an edge of N_{j-2}. Note that we cannot have $(x, y) = (v_{2k+2}, v_{2k+1})$ because N_{j-1} is deceptive, so v_{2k+2} and v_1 are not adjacent. We handle the case $(x, y) = (v_{2k+4}, v_{2k+3})$ separately, and the other cases all at once by defining an the alternating path P of length 5, going from v_1 to v_{2k+l} for some $l \in \{1, 2\}$, that we will use later to define the appropriate switch:

Case 3 If $(x, y) = (v_{2k+4}, v_{2k+3})$, then since $yv_{2k} = v_{2k+3}v_{2k}$ and $v_{2k+1}v_{2k+2}$ are edges of N_{j-1}, by performing the 2-switch $v_{2k+3}v_{2k}v_{2k+1}v_{2k+2}$, one obtains a matching the representative matching N_j with progress index equal to $k + 2$.

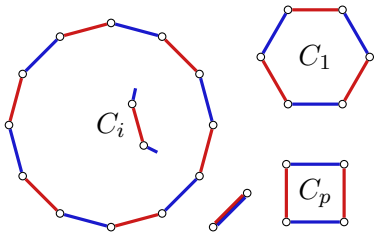
Case 4 If $(x, y) = (v_{2k+3}, v_{2k+4})$, let $P = v_1v_{2k+3}v_{2k+2}v_{2k+1}v_{2k}v_{2k+4}$ and $l = 2$. Note that this case cannot occur if G is bipartite because v_1 and v_{2k+3} are then in the same half of the bipartition.

Case 5 If $(x, y) = (v_{2k+1}, v_{2k+2})$, let $P = v_1v_{2k+1}v_{2k}v_{2k+2}v_{2k+3}v_{2k+4}$ and $l = 2$. Note again that this case cannot occur if G is bipartite because v_1 and v_{2k+1} are then in the same half of the bipartition.

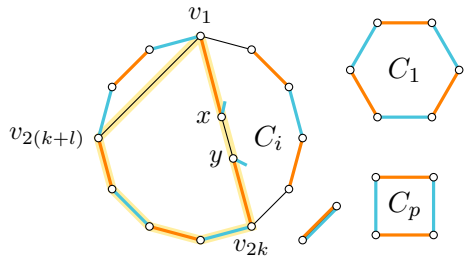
Case 6 Otherwise, we are in the generic case $\{x, y\} \cap \{v_{2k+1}, \dots v_{2k+4}\}$. Let $P = v_1xyv_{2k} \dots v_{2k+2}$ and $l = 1$.

As Case 3 is already settled, we assume from now on that $(x, y) \neq (v_{2k+4}, v_{2k+3})$. We handle Cases 4, 5 and 6 simultaneously. In each of these cases, one can check that P is an alternating path of N_{j-1}. We distinguish two subcases, depending on whether there exists some $m \in \{0, 1\}$ such that $v_{2(k+l+m)}$ is a neighbour of v_1.

Cases 4a, 5a and 6a Assume that there exists some $m \in \{0, 1\}$ such that $v_{2(k+l+m)}$ is a neighbour of v_1, and let m be the largest such integer. Let C be the alternating cycle obtained by concatenating P with $v_{2(k+l)} \dots v_{2(k+l+m)} v_1$. The cycle C has length six or eight because P hes length five. Let N_j be the perfect matching obtained by performing the 4-switch on C (see for example Figure 11 for Case 6a, with $P = v_1 x y v_{2k} \dots v_{2k+4}$ and $l = 2$ and $m = 0$). The vertices affected by this 4-switch all belong to $C_i \cup \{x, y\}$. If we had $(x, y) = (v_{2k+3}, v_{2k+4})$ or $(x, y) = (v_{2k+1}, v_{2k+2})$ (that is Cases 4a and 5a respectively), N_j is representative with progress index $k+l+m$: it is equal to M_i on all vertices but $\{v_1, \dots v_{2(k+l+m)}\}$, where N_j contains $v_1 v_{2(k+l+m)}$ and is equal to M_{i+1} on all other vertices. In Case 6a, recall that $P = v_1 x y v_{2k} \dots v_{2k+2}$. Since N_{j-1} was misleading, xy belonged to N_{j-2} and belongs to N_j as well because x and y are consecutive in P. Thus, N_j is representative with progress index $k+l+m$: it is equal to M_i on all vertices but $\{v_1, \dots v_{2(k+l+m)}\}$, where N_j contains $v_1 v_{2(k+l+m)}$ and is equal to M_{i+1} on all other vertices.



(a) The perfect matchings S, and T are represented in blue and red respectively.



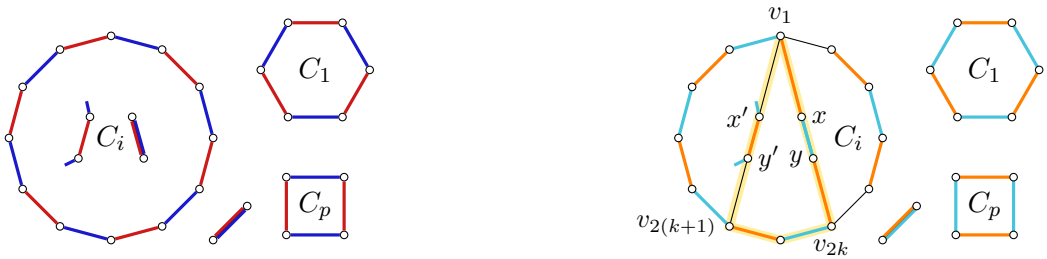
(b) The matchings N_{j-1} and L (defined in Claim 26), represented in orange and light blue respectively. N_j is obtained by performing a 4-switch on C, which is here highlighted in yellow.

Figure 11: Constructing N_j when N_{j-1} is misleading and v_1 is adjacent to $v_{2(k+l)}$ for some $l \in [2]$ (here $l = 2$). The resulting N_j is representative.

Cases 4b, 5b, and 6b Finally, assume that no such m exists. Let $V' = V(G) \setminus V(P)$ and A be the set of vertices of V' matched in N_{j-1} to a neighbour of v_1[5]. Since v_1 is not adjacent to v_{2k} and $v_{2(k+l)}$, it has at least $\deg(v_1) - 3$ neighbours among V', so $|A| \geq \deg(v_1) - 3$. Likewise, $v_{2(k+l)}$ has at least $\deg(v_{2(k+l)}) - 4$ neighbours in V' because $v_{2(k+l)}$ and v_1 are not adjacent. So $|A| + |B| \geq n - 5 > |V'|$ and by pigeonhole principle A and B intersect. In other words, there exists $x'y' \in N_{j-1}$, with $x', y' \notin V(P)$, such that x' is a neighbour of v_1 and y' of $v_{2(k+l)}$. Let C be the cycle of length eight obtained by concatenating P with $v_{2(k+l)} y' x' v_1$. By performing the 4-switch on the C in N_{j-1}, one obtains the misleading matching N_j with the deceptive edge $x'y'$ and progress index $k + l$ (see for example Figure 12 for Case 6b where $l = 1$ and $P = v_1 x y v_{2k} \dots v_{2k+2}$). Indeed, if we had $(x, y) = (v_{2k+1}, v_{2k+2})$ or $(x, y) = (x, y) = (v_{2k+3}, v_{2k+4})$ (that is Cases 4b and 5b respectively), then recall that $P = v_1 v_{2k+1} v_{2k} v_{2k+2} v_{2k+3} v_{2k+4}$ and $v_1 v_{2k+3} v_{2k+2} v_{2k+1} v_{2k} v_{2k+4}$ respectively, so N_j is equal to M_i on all vertices but $\{v_1, \dots v_{2(k+2)}, x', y'\}$, where N_j contains $v_1 v_{2k+2}$ and $x'y'$, and is equal to M_{i+1} on all other vertices. Since m did not exists, neither

[5]If G is a balanced bipartite graph, recall that $P = v_1 x y v_{2k} \dots v_{2k+2}$ and $l = 1$. Let $V' = V_1 \setminus \{v_1, y, v_{2k+1}\}$. We have $|A| \geq \deg(v_1) - 1$ because v_1 is not adjacent to v_{2k} or v_{2k+2}. Let $B = N(v_{2k+2} \cap V')$, we have $|B| \geq \deg(v_{2k+2}) - 2$ because v_1 is not adjacent to v_{2k+2}. By assumption, $|A| + |B| \geq n - 2$ so by pigeonhole principle, since $|V'| = n - 3$, A and B intersect.

v_{2k+4} nor v_{2k+6} are adjacent to v_1. In the remaining case, recall that $P = v_1xyv_{2k}\dots v_{2k+2}$. Since N_{j-1} was misleading with deceptive edge xy, the edge xy belonged to N_{j-2} and also belongs to N_j because x and y are consecutive in P. Thus N_j is equal to M_i on all vertices but $\{v_1, \dots v_{2(k+2)}, x', y'\}$, where N_j contains v_1v_{2k+2} and $x'y'$, and is equal to M_{i+1} on all other vertices. Finally, note that by assumption, since m did not exists, neither v_{2k+2} nor v_{2k+4} are adjacent to v_1. ■



(a) The perfect matchings S, and T are represented in blue and red respectively.

(b) The matchings N_{j-1} and L (defined in Claim 26), represented in orange and light blue respectively. N_j is obtained by performing a 4-switch on C, which is here highlighted in yellow.

Figure 12: Constructing N_j when N_{j-1} is misleading and v_1 has no neighbours among $v_{2(k+1)}$ and $v_{2(k+2)}$. The resulting N_j is misleading.

Note that the progress index increases at each step of our reconfiguration sequence, so the sequence between M_i and M_{i+1} has length at most $|C_i| = |M_i \Delta M_{i+1}|$. So the canonical path between S and T has length at most n.

Analysis of the congestion

Given some transition $M \to M'$, let $\mathrm{cp}(M, M')$ denote the set of couple (S, T) such that the canonical path between S and T includes the transition $M \to M'$. Recall that the congestion of our set of canonical paths is defined by

$$\varrho(\Gamma) = \max_{M \to M'} \left\{ \frac{\sum_{(S,T) \in \mathrm{cp}(M,M')} \pi(S)\pi(T)|\gamma_{S,T}|}{\pi(M)\,\mathrm{Prob}(M \to M')} \right\} \qquad \text{where } \pi \text{ is the uniform distribution on perfect matchings}$$

$$\leq n \max_{M \to M'} \frac{\pi(M)|cp(M, M')|}{\mathrm{Prob}(M \to M')} \qquad \text{Because } |\gamma_{S,T}| \leq n$$

Thus we need to bound the number of canonical paths using a transition $M \to M'$. Given some $c \in \{1, 2, 3, 4a, 4b, 5a, 5b, 6a, 6b\}$, denote $\mathrm{cp}_c(M, M')$ the couples $(S, T) \in \mathrm{cp}(M, M')$ such that in the canonical path from S to T, M' is constructed using Case c of Claim 25.

Claim 26. *For any fixed c and transition $M \to M'$, we have*

$$\frac{|\,\mathrm{cp}_c(M, M')|}{\mathrm{Prob}(M, M')} = O(\pi(M)^{-1}n^6).$$

Proof of Claim. Let $c \in \{1, 2, 3, 4a, 4b, 5a, 5b, 6a, 6b\}$ and $M \to M'$ be a fixed transition and $C = M \Delta M'$ the cycle on which the 4-switch is performed. We define an injective application on $\mathrm{cp}_c(M, M')$ to count it. The Case 3 is somewhat different from the others, because it is the only case where v_1 does not belong to the switch that is performed. For this reason, the definition of our injective application differs slightly for $c = 3$. If $c = 3$, let f_3 be the application that associates to any $(S, T) \in \mathrm{cp}_c(M, M')$ a tuple (L, a, v, w) where L is perfect or near-perfect

matching with some additional information $a \in \{0,1\}$ and $v, w \in V(G)$ to be defined later. Otherwise, let f_c be the application that associates to any $(S,T) \in \mathrm{cp}_c(M,M')$ a tuple (L,a,b), where L is again a perfect or near-perfect matching, with some additional information $a \in \{0,1\}$ and $b \in [8]$ to be defined later.

Let $(S,T) \in \mathrm{cp}(M,M')$. Recall that $C_1, \dots C_p$ are the alternating cycles in $S\Delta T$ ordered by smallest vertex and that for all i, M_i is the perfect matching that is equal to T on all the cycles C_j with $j < i$ and to S on all the remaining vertices. Let i such that $M \to M'$, was used when connecting M_i to M_{i+1} and denote again $v_1, v_2, \dots v_{2q}$ the consecutive vertices of C_i, such that v_1 is the smallest vertex and $v_1v_2 \in M_i$. Let k and k' be the progress indices of M and M' respectively. Let L be the matching defined by

$$L := (S \cap T) \ \cup \ \left(S \cap (\bigcup_{j<i} C_j) \right) \ \cup \ \left(T \cap (\bigcup_{j>i} C_j) \right)$$
$$\cup \ \{v_{2i+1}v_{2i+2} : 1 \le i \le k-2\} \ \cup \ \{v_{2i}v_{2i+1} : k \le i \le q\}.$$

In other words, L is equal to S on all vertices in $\bigcup_{j<i} V(C_j) \cup \{v_3, \dots v_{2k-2}\}$, to T on all vertices in $\bigcup_{j>i} V(C_j) \cup \{v_{2k}, \dots v_{2q}, v_1\}$ and to S (and T) on all vertices of $V(G) \setminus \bigcup_{j=1}^{p} V(C_j)$. Note that L is a perfect matching if and only if $k = 1$ that is if $M = M_i$. Otherwise, L is a near-perfect matching with unmatched vertices v_2 and v_{2k-1}.

Let a denote whether v_2 is greater than v_{2k-1}. If $c = 3$, then v_1 does not belong to the cycle $C = v_{2k}v_{2k+1}v_{2k+2}v_{2k+3}$ that is switched and let $v = v_1$ and $w = v_{2k}$. In all other cases, v_1 belongs to C and we set b to denote the rank within the order of v_1 among the (at most) eight vertices of C.

We will now prove that f_c is injective. Let $(L,a,v) = f_3(S,T)$ or $(L,a,b) = f_c(S,T)$ for some $(S,T) \in \mathrm{cp}(M,M')$. The support of the 4-switch can be recovered: $C = M\Delta M'$. The vertex v_1 can also be recovered: if $c = 3$ it is v, if $c \neq 3$ then it has rank b within the order of the vertices of C.

We now prove that v_2 and the alternating cycles in $S\Delta T$ can be recovered as well. First assume that L is perfect. Then $M = M_i$, so M (respectively L) is equal to T (respectively S) on all vertices in $\bigcup_{j<i} V(C_j)$, to S (respectively T) on all vertices in $\bigcup_{j\ge i} V(C_j)$ and to $S \cap T$) on all other vertices. In other words $L\Delta M = S\Delta T$ and we have recovered the set of alternating cycles. The vertex v_2 is matched to v_1 in M, so we can recover v_2 as well. Thus we can assume that L is not perfect. Let $N = L$ if $c \neq 3$ and $N = L \setminus \{v_1v_{2k+4}\}$ if $c = 3$ (this is well defined as we have already recovered v_1 and its neighbour in L is v_{2k+4}). Consider $H = N\Delta(M \cap M')$, it is a collection of alternating paths and cycles. Note that the endpoints of the alternating paths of H are the unmatched vertices of N and $M \cap M'$, that is $A = V(C) \cup \{v_2, v_{2k-1}\}$ if $c \neq 3$ and $A = V(C) \cup \{v_2, v_{2k-1}, v_1, v_{2k+4}\}$ if $c = 3$. The matching N is equal to S on $\bigcup_{j<i} V(C_j) \cup \{v_2, \dots v_{2k-1}\} \setminus A$, to T on $\bigcup_{j<i} V(C_j) \cup \{v_{2k+1}, \dots v_{2q}\} \setminus A$, and to $S \cap T$ on all remaining vertices of $V(G) \setminus A$. On the other hand, regardless of whether M and M' were representative or misleading in the sequence going from S to T, they were both equal to T on $\bigcup_{j<i} V(C_j) \cup \{v_2, \dots v_{2k-1}\} \setminus A$, to S on $\bigcup_{j<i} V(C_j) \cup \{v_{2k+1}, \dots v_{2q}\} \setminus A$, and to $S \cap T$ on all remaining vertices of $V(G) \setminus A$. So all alternating cycles in H are alternating cycles of $S\Delta T$ and the alternating paths in H are subpaths of the alternating paths of $S\Delta T$. By identifying the labels of the vertices in A, the value of c will then be enough information to recover how to combine the alternating paths in H into alternating cycles of $S\Delta T$. Recall that $A = V(C) \cup \{v_2, v_{2k-1}\}$ if $c \neq 3$ and $A = V(C) \cup \{v_2, v_{2k-1}, v_1, v_{2k+4}\}$ if $c = 3$. The vertices v_2 and v_{2k-1} can be recovered using a and the fact that v_2 and v_{2k-1} are the only unmatched vertices of L. If $c \neq 3$, then the labels of $V(C)$ are fully determined by v_1, $M \cap C$ and $M' \cap C$. If $c = 3$, the labels of $V(C)$ are fully determined by $M \cap C$, $M' \cap C$ and the value of $w = v_{2k} \in V(C)$.

Now that we have recovered the alternating cycles in $S\Delta T$, their numbering can be recovered

using the ordering on the vertices. S is equal to L on all cycles C_j with $j < i$ and equal to $C_j \Delta L$ on all cycles with $j > i$ (and vice versa for T). Regarding C_i, S and T alternate on it and S contains the edge $v_1 v_2$, which completes the description of S and T.

By Lemma 7, the number of values that f_c can take is $O(n^2 \pi(M)^{-1})$, so there are $O(n^2 \pi(M)^{-1})$ canonical path in $\mathrm{cp}_c(M, M')$ for $c \neq 3$ and $O(n^4 \pi(M)^{-1})$ canonical paths in $\mathrm{cp}_c(M, M')$ for $c = 3$. If $c = 3$, then the switch $M \to M'$ is a 2-switch, so $\mathrm{Prob}(M \to M') = \Theta(1/n^2)$. On the other hand $\mathrm{Prob}(M \to M') = \Omega(1/n^4)$ for all 4-switches. So for all c,

$$\frac{|\mathrm{cp}_c(M, M')|}{\mathrm{Prob}(M, M')} = O(\pi(M)^{-1} n^6). \qquad \blacksquare$$

It follows directly that

$$\varrho(\Gamma) \leq n \max_{M \to M'} \frac{\pi(M) |cp(M, M')|}{\mathrm{Prob}(M \to M')} = O(n^7)$$

As the number of perfect matching of G is $\pi(M)^{-1} \leq n!/(n/2)!$, we have

$$\tau_{\mathrm{mix}}(\Gamma) = O(\varrho(\Gamma) \log(\pi(M)^{-1})) = O(n^8 \log(n)). \qquad \square$$

5.2 Polynomial mixing time of the 4-switch Markov chain

A direct consequence of Theorem 8 and Theorem 9, is that the 2-switch and 3-switch Markov chains also have polynomial mixing times in these respective regimes, which implies Theorem 6 that we recall:

Theorem 6. *Let G be an n-vertex graph with $\delta(G) \geq n/2+2$ (or respectively $\delta(G) \geq \lfloor 2n/3 \rfloor + 1$). Alternatively, let G be a balanced bipartite graph on $2n$ vertices with $\delta(G) \geq \lfloor n/2 \rfloor + 1$ (or respectively $\delta(G) \geq \lfloor 2n/3 \rfloor + 1$).*
The random walk on $\mathcal{H}_3(G)$ (respectively $\mathcal{H}_2(G)$) has mixing time polynomial in n.

Proof. By Theorem 8, 4-switches can be realised by a sequence of length $O(1)$ of 3-switches if $\delta(G) \geq n/2 + 2$ (alternatively $\delta(G) \geq \lfloor n/2 \rfloor + 1$ if G is balanced bipartite). Likewise, 3-switches can be realised by sequences of length $O(1)$ of 2-switches if $\delta(G) \geq \lfloor 2n/3 \rfloor + 1$). For each 4-switch $(M \to M')$, define the canonical path between M and M' to be an arbitrary such sequence. The number of vertices involved in this sequence is bounded by a constant c, thus the number of canonical paths using any fixed 3-switch or 2-switch is bounded by $n^{O(1)}$. By Markov chain comparison, this proves polynomial mixing time for the random walk on $\mathcal{H}_3(G)$ (respectively on $\mathcal{H}_2(G)$) in the aforementioned regimes. $\square$

6 Isolated matchings

The celebrated Caccetta-Häggkvist conjecture [13] asserts that every oriented n-vertex graph with minimum outdegree at least n/k contains a directed cycle of length at most k. The weaker conjecture obtained by replacing the minimum outdegree condition by having minimum semidegree (minimum of outdegrees and indegrees over all vertices) at least n/k is also still open. The following two theorems show how each of these conjectures is related to the thresholds $\delta_k^{\mathbf{noiso}}$ and $\delta_{k,\mathrm{bip}}^{\mathbf{noiso}}$ of appearance of isolated vertices in the k-switch graph:

Theorem 27. *Let $k \geq 2$ and n be an even integer. Let $d_k(n)$ be the largest integer such that there exists an n-vertex graph with minimum* outdegree *$d_k(n)$ and no directed cycle of length at most k. Then,*

$$\delta_k^{\mathbf{noiso}}(n) \leq d_k(n) + 2.$$

Theorem 28. *For all $k \geq 2$, all d and all n, there exists an oriented n-vertex graph with minimum* semidegree *d and no directed cycle of length at most k if and only if there exists a bipartite graph G on $2n$ vertices with $\delta(G) = d+1$ such that $\mathcal{H}_k(G)$ has an isolated vertex.*

Proof of Theorem 27. Let G be an n-vertex graph with a perfect matching M that is an isolated vertex of $\mathcal{H}_k(G)$. Let δ be the minimum degree of G. Denote $V(G) = \{u_1, \dots, u_{n/2}\} \cup \{v_1, \dots, v_{n/2}\}$, where u_i is matched to v_i in M for each $i \in [n/2]$. For convenience, we will denote $\bar{x}$ the vertex to which a vertex x is matched to in M. Let D be the oriented graph on the same vertex set $\{u_1, \dots, u_{n/2}\} \cup \{v_1, \dots, v_{n/2}\}$ defined as follows. For each $i \in [n/2]$, for each vertex $w \in V(D) \setminus \{u_i, v_i\}$, add an arc from u_i to w if v_i and w are adjacent in G, and symmetrically an arc from v_i to w if u_i and w are adjacent in G. In other words, we have $E(D) = \{\bar{x}w \colon xw \in E(G) \setminus E(M)\}$. Let $C = x_1 \dots x_p$ be a directed cycle of D, we will show that $p \geq k+1$. For each $i \in [p]$, $x_i\bar{x}_i$ is by definition an edge of M, and by construction, $\bar{x}_i$ and $x_{i+1 \pmod p}$ are adjacent in G. So $x_1\bar{x}_1 \dots x_p\bar{x}_p$ is an alternating cycle in G, which contradicts the assumption that M is an isolated vertex of $\mathcal{H}_k(G)$ if $p \leq k$. Hence D has no directed cycles of length at most k, and in particular, D is oriented. The outdegree of each vertex $x \in V(D)$ is equal to $\deg_G(\bar{x}) - 1$, so the digraph D has minimum outdegree equal to $\delta(G) - 1$. So $\delta(G) \leq d_k(n) + 1$ for all n-vertex graphs G with an isolated matching in $\mathcal{H}_k(G)$. Hence $\delta_k^{\mathbf{noiso}}(n) \leq d_k(n) + 2$. □

Proof of Theorem 28. We first prove the direct implication. Let D be an oriented n-vertex graph with minimum semidegree d and no directed cycle of length at most k. Let us denote $V(D) = \{u_1, \dots, u_n\}$. Let G be the graph on the vertex set $\{a_1, \dots a_n\} \cup \{b_1, \dots b_n\}$ constructed as follows. For each i, connect a_i to b_i. For each distinct i and j, connect a_i to b_j if $u_i \to u_j \in E(D)$. The graph G has minimum degree $d+1$. Let M be the matching $\{a_ib_i \colon i \in [n]\}$. A k-switch on M is a cycle $a_{i_1}b_{i_1} \dots a_{i_\ell}b_{i_\ell}$ of length 2ℓ for some $\ell \leq k$. But no such cycle exists as it would imply that $u_{i_1} \dots u_{i_\ell}$ is a directed cycle of length at most k in D, so M is an isolated vertex of $\mathcal{H}_2(D)$.

Conversely, let G be a balanced bipartite graph on $2n$ vertices with $\delta(G) = d+1$ and an isolated matching $M = \{a_ib_i \colon i \in [n]\}$ in $\mathcal{H}_k(G)$. Let D be the oriented n-vertex graph constructed as follows. Denote $V(D) = \{u_1, \dots u_n\}$ and for all distinct i and j, add the arc $u_i \to u_j$ if $a_ib_j \in E(G)$. The minimum semidegree of D is d. A directed cycle $u_{i_1} \dots u_{i\ell}$ in D corresponds to the cycle $a_{i_1}b_{i_1} \dots a_{i_\ell}b_{i_\ell}$. Since M is isolated in $\mathcal{H}_k(G)$, such cycle has length at least $2(k+1)$, so D contains no directed cycle of length at most k (in particular D is oriented because $k \geq 2$. □

This suggests that $\delta_{k,\text{bip}}^{\mathbf{noiso}}(n) = \lceil n/k \rceil + 1$ and since the bounds we obtained in the balanced bipartite case and the general case were all equal up to $O(1)$, we conjecture the following:

Conjecture 1. *There exists $c > 0$ such that for all k and n,*

$$\delta_k^{\mathbf{noiso}}(n) \in [n/k - c, n/k + c].$$

This conjecture holds for $k = 2$, as we obtain a sharp threshold for the appearance of an isolated matching (in balanced bipartite graphs and general graphs):

Theorem 29. *Let G be n-vertex graph (or respectively a balanced bipartite graph on $2n$ vertices). If $\delta(G) \geq n/2 + 1$ (respectively $\delta(G) \geq \lfloor (n+3)/2 \rfloor$), then for every perfect matching M of G and every $e \in E(M)$, there exists a 2-switch that contains e.*
On the other hand, $\mathcal{H}_2(G)$ may have isolated vertices if $\delta(G) \leq n/2 - 2$ (respectively $\lfloor (n+1)/2 \rfloor$). In other words,

$$\frac{n}{2} - 1 \leq \delta_2^{\mathbf{noiso}}(n) \leq \frac{n}{2} + 1 \quad \text{and} \quad \delta_{2,\text{bip}}^{\mathbf{noiso}}(n) = \left\lfloor \frac{n+3}{2} \right\rfloor.$$

Proof. We first prove the existence of such a switch the minimum degree is sufficiently high. Let M be a perfect matching of G and uv be an edge of M. Let $V' = V(G) \setminus \{u, v\}$, A be the subset of vertices of V' that are matched in M to a neighbour of u, and $B = N(v) \cap V'$ [6]. We have $|A| \geq \deg(u) - 1 \geq n/2$ and $|B| \geq n/2$. By pigeonhole principle, A and B intersect, so there exists a 2-switch that contains uv.

Conversely, we construct graphs with high minimum degree and an isolated matching. To do so, we first construct an auxiliary oriented graph H_p on p vertices with maximal semidegree $\delta(H_p) := \max_{u_i} \max\{\delta^+(u_i), \delta^-(u_i)\}$. The semidegree of an oriented graph on p vertices is at most $\lfloor (p-1)/2 \rfloor$. This bound is attained by taking H_p to be the oriented graph such that there is an arc from u_i to u_j is $j - i \in \{1, \ldots, \lfloor (p-1)/2 \rfloor\} \pmod p$. By Theorem 27, there exists a balanced bipartite graph $G_n^{(\mathcal{B})}$ on $2n$ vertices with minimum degree $\lfloor (n-1)/2 \rfloor + 1 = \lfloor (n+1)/2 \rfloor$, such that $\mathcal{H}_2(G_n^{(\mathcal{B})})$ contains an isolated vertex.

We now give the construction for general graphs. Let n be an even integer and $p = n/2$ and consider the graph G_n on the vertex set $\{a_1, \ldots a_p\} \cup \{b_1, \ldots b_p\}$ built from H_p as follows. For each i, connect a_i with b_i. For each vertex $u_i \in V(H_p)$, we partition $N^+(u_i)$ into $A_i \cup B_i$ such that $||A_i| - |B_i|| \leq 1$. We connect a_i to a_j and b_j for all $u_j \in A_i$, and b_i to a_j and b_j for all $u_j \in B_i$. The graph G_n constructed this way has minimum degree equal to $1 + 2\lfloor \delta^+(H_p)/2 \rfloor + \delta^-(H_p)$. Moreover, the matching $M = \{a_i b_i \colon i \in [n/2]\}$ is isolated in $\mathcal{H}_2(G_n)$: since H_p is an oriented graph, for each distinct i and j, the subgraph of G_n induced by $\{a_i, b_i, a_j, b_j\}$ is isomorphic to a triangle with an edge attached to one of the vertices, so the edges $a_i b_i$ and $a_j b_j$ cannot be part of a 2-switch. The minimum degree of G_n is

$$\begin{aligned}\delta(G_n) &= 1 + 2\left\lfloor \frac{p-1}{4} \right\rfloor + \left\lfloor \frac{p-1}{2} \right\rfloor = 2\left\lceil \frac{p}{4} \right\rceil + \left\lceil \frac{p}{2} \right\rceil - 2 \\ &= 2\left\lceil \frac{n}{8} \right\rceil + \left\lceil \frac{n}{4} \right\rceil - 2 \geq n/2 - 2.\end{aligned}$$

□

Note that the first statement of Theorem 29 in particular implies that no edge of M is frozen, so

$$\delta_k^{\mathbf{thaw}}(n) \leq \frac{n}{2} + 1 \quad \text{and} \quad \delta_{k,\mathrm{bip}}^{\mathbf{thaw}}(n) \leq \left\lfloor \frac{n+3}{2} \right\rfloor.$$

7 Open questions and perspectives

7.1 Perfect matchings

We highlight three questions about geometry in the space of perfect matchings. What happens for regular balanced bipartite graphs? What is the threshold for the appearance of a giant component in the 2-switch graph? What is the threshold guaranteeing no isolated vertices in the k-switch graph for $k \geq 3$?

Regular graphs For any d, the number $\phi(G)$ of perfect matching of any d-regular balanced bipartite graph G is exponential [39, 42, 24, 15]:

$$\Phi(G) \geq \left(\frac{(d-1)^{d-1}}{d^{d-2}} \right)^n.$$

The "volume" heuristic suggests that for all d, k and n, and any d-regular balanced bipartite graph G on $2n$ vertices, $\mathcal{H}_k(G)$ is connected. This is false as for all d, there exists d-regular balanced bipartite graphs with arbitrarily large girth. This is witnessed by a uniform random

[6] If G is a balanced bipartite graph, then let $V' = U \setminus \{u\}$, where U is the half of the bipartition containing u, and let A and B be defined identically. Then A and B have size at least $\lfloor (n+1)/2 \rfloor$, so $|A| + |B| \geq n > |V'|$ and by pigeon hole principle, A and B intersect.

d-regular graph, but also by blowing-up of the vertices of a $2(k+1)$-cycle into independent sets of equal size. The resulting graph has a disconnected k-switch graph (see Lemma 22) and degree $\frac{2n}{k+1}$, where $2n$ is the number of vertices. We conjecture that this blow-up maximises $\delta(G)$ among regular balanced bipartite graphs on $2n$ vertices with disconnected k-switch graph:

Conjecture 2. *There exists $c \geq 0$ such that for all n and k,*

$$\delta^{\mathbf{connect}}_{k,\mathrm{reg}}(n) \in \left[\frac{2n}{k+1} - c, \frac{2n}{k+1} + c\right].$$

Note that Theorem 9 and Theorem 8 confirm the upper bound of this conjecture for $k = 2$ and $k = 3$, and that Corollary 23 confirms the lower bound when $k+1$ divides n. More generally, we believe that the switch graph of d-regular balanced bipartite graphs on $2n$ vertices undergoes a phase transition around $2n/(k+1)$ that is similar to that described in this article for balanced bipartite graphs and general graphs.

Giant component Theorem 9 and Corollary 19 place the threshold for the appearance of a giant component in the 2-switch graph between $n/2 - \varepsilon$ and $\lfloor (2n+3)/3 \rfloor$. We conjecture that the giant component threshold is near the lower bound.

Conjecture 3. *For each $c \geq 1$, there exists $\varepsilon \geq 0$ such that for all n, for all n-vertex graphs G (or alternatively all balanced bipartite graphs on $2n$ vertices) with minimum degree at least $n/2 + \varepsilon$, the 2-switch graph $\mathcal{H}_2(G)$ contains a component of size $|\mathcal{H}_2(G)|/c$.*

Isolated matchings For isolated matchings, we have seen in Section 6 that the conjecture of Caccetta and Häggkvist about the maximal minimum outdegree of an oriented graph without directed cycles of length k would directly imply, if true, an upper bound on the threshold $\delta^{\mathbf{noiso}}_k(n)$ at which the k-switch graph is guaranteed to have positive minimum degree. A weaker version of this conjecture, in which outdegree is replaced by semidegree, is actually equivalent to the minimum degree threshold that guarantees no isolated vertices in the k-switch graph of balanced bipartite graphs. This suggests that for all k and n, $\delta^{\mathbf{noiso}}_{k,\mathrm{bip}}(n) = \lceil n/k \rceil + 1$ and motivates the following conjecture for general graphs:

Conjecture 1. *There exists $c > 0$ such that for all k and n,*

$$\delta^{\mathbf{noiso}}_k(n) \in [n/k - c, n/k + c].$$

7.2 Phase transitions for other Dirac-type theorems

Many other structures exhibit phase transitions close to the minimum degree at which the structure is guaranteed to exist in G. Most of the existing literature expresses these phase transitions in one of the following terms. Above the non-emptiness threshold, there are $\Omega(n)^n$ embeddings of the structure. The host graph contains the structure in a robust sense: sparsifying the host graph with probability $\Omega(\log n/n)$ preserves the existence of an embedding of the structure (see the survey of Sudakov [40]), thereby witnessing that these embeddings are "everywhere" in the host graph. More recently, this phase transition was also described by the existence of spread distributions, which in particular imply the counting and robustness results we just mentioned.

For example, these depictions of the phase transition were described for Hamiltonian cycles (see [38, 16, 17] for an estimate of the number of Hamiltonian cycles, [32] for the robustness and [29] for the existence of spread distributions), triangle factors or more generally clique factors (see [4, 3] for the robust versions of the Corrádi and Hajnal theorem and [36, 29] for the existence of spread distributions) and spanning trees of bounded degree (see [31] for an counting estimate and [36, 5] for spread distributions).

However, except for Hamiltonian cycles [30] and the present work for perfect matchings, the phase transition has not been described in terms of the geometry of the configuration space. Given a spanning structure $(S_n)_{n\in\mathbb{N}}$, for any n-vertex graph G and integer k, let $\mathcal{H}_k(G)$ be the reconfiguration graph of the embeddings of S_n, where two copies of S_n in G are adjacent if they differ by at most k edges. Motivated by our results for perfect matchings and by the results of [30] for the connectedness threshold of Hamiltonian cycles, we cast the following problem.

Problem 30. For any spanning structure $(S_n)_{n\in\mathbb{N}}$ with non-emptiness threshold δ_n, does there exists an integer $k = O(1)$ such that the geometry of $\mathcal{H}_k(G)$ undergoes a phase transition close to δ_n?

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